

CM RetailOS Suite v3.0 User Guide

The definitive illustrated operational handbook for cashiers, stock managers, accountants, and store owners. Master first-time setup, hardware licensing, split-second barcode checkouts, inventory tracking, shift reconciliation, offline profit intelligence, and standalone spreadsheet auditing.

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First-Time Setup & Onboarding Wizard

When launching CM RetailOS for the first time, an intuitive 4-step wizard personalizes your workspace theme, currency, store branding, and root administrator credentials.

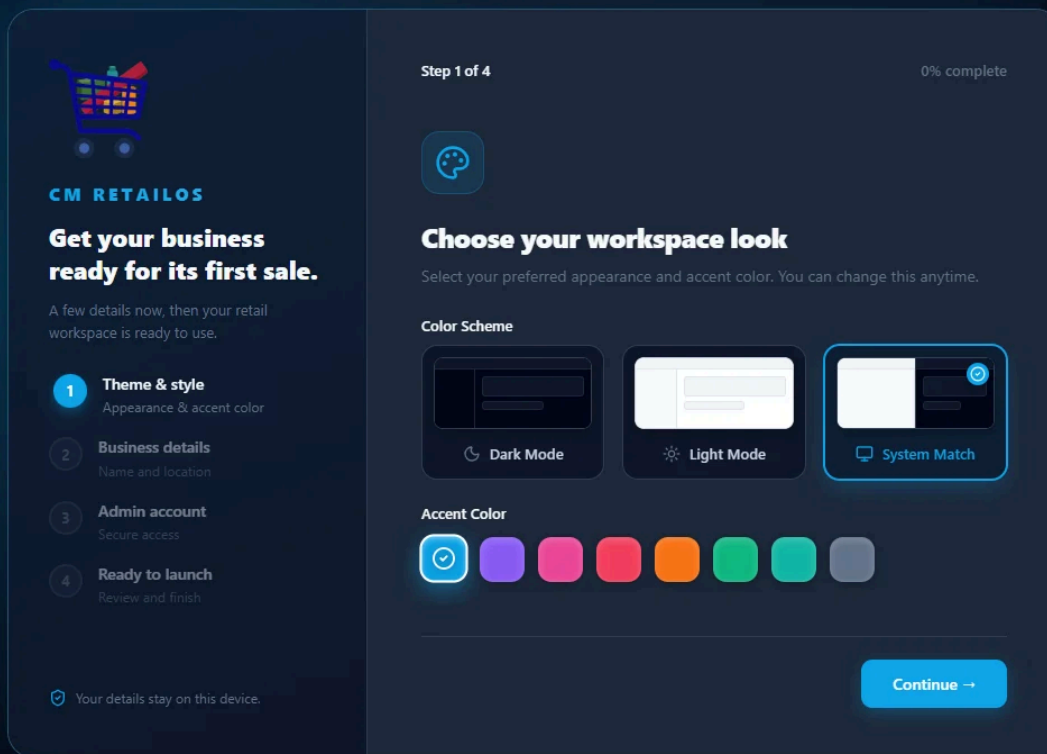


FIGURE 1.1

Setup Wizard — Step 1: Color Scheme (Dark Mode, Light Mode, System Match) & 8-Color Accent Palette

HOW TO READ & USE THIS SCREEN

This is the very first screen CM RetailOS shows after installation — Step 1 of 4 of the onboarding wizard. The left rail tracks your progress through Theme & style → Business details → Admin account → Ready to launch, while the main panel asks you to choose your workspace look: a Color Scheme (Dark Mode, Light Mode or System Match) and an Accent Color from the 8-swatch palette.

- 1 Pick a Color Scheme — Dark Mode is easiest on the eyes for night shifts, Light Mode suits bright sunlit shops, and System Match simply follows your Windows setting.
- 2 Click any accent swatch to set the highlight color used on buttons, active menus and charts across the whole app.
- 3 Press Continue to move on to Step 2.

 **Pro tip:** This choice is purely cosmetic — you can change the scheme and accent any time later under Settings → Appearance & Theme.

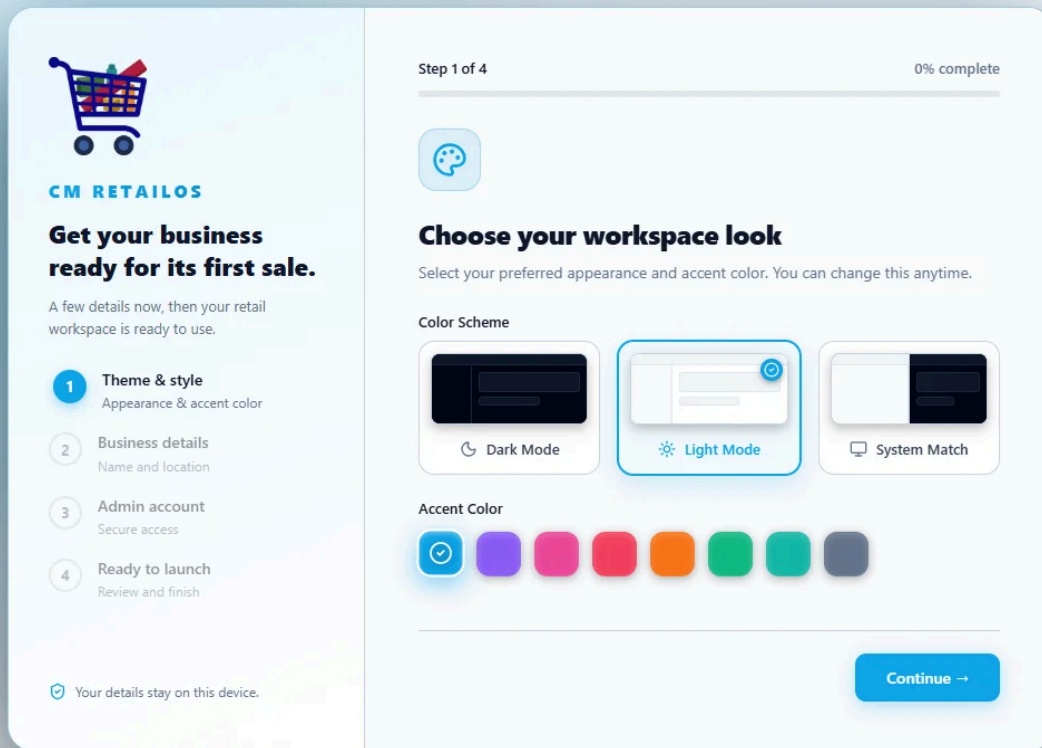


FIGURE 1.2 Step 1: High-Visibility Light Mode Active

HOW TO READ & USE THIS SCREEN

The same Step 1 with Light Mode selected so you can compare both looks before committing. Notice the two reassurances on this screen: “Select your preferred appearance and accent color. You can change this anytime.” and “Your details stay on this device.” — the wizard saves everything locally; no internet account is created.

- 1 Toggle between Dark and Light and watch the entire wizard re-skin instantly.
- 2 Judge readability from your real counter distance — sunlight near windows washes out dark themes.
- 3 Confirm your choice and continue; the progress bar reads Step 1 of 4, 0% complete.

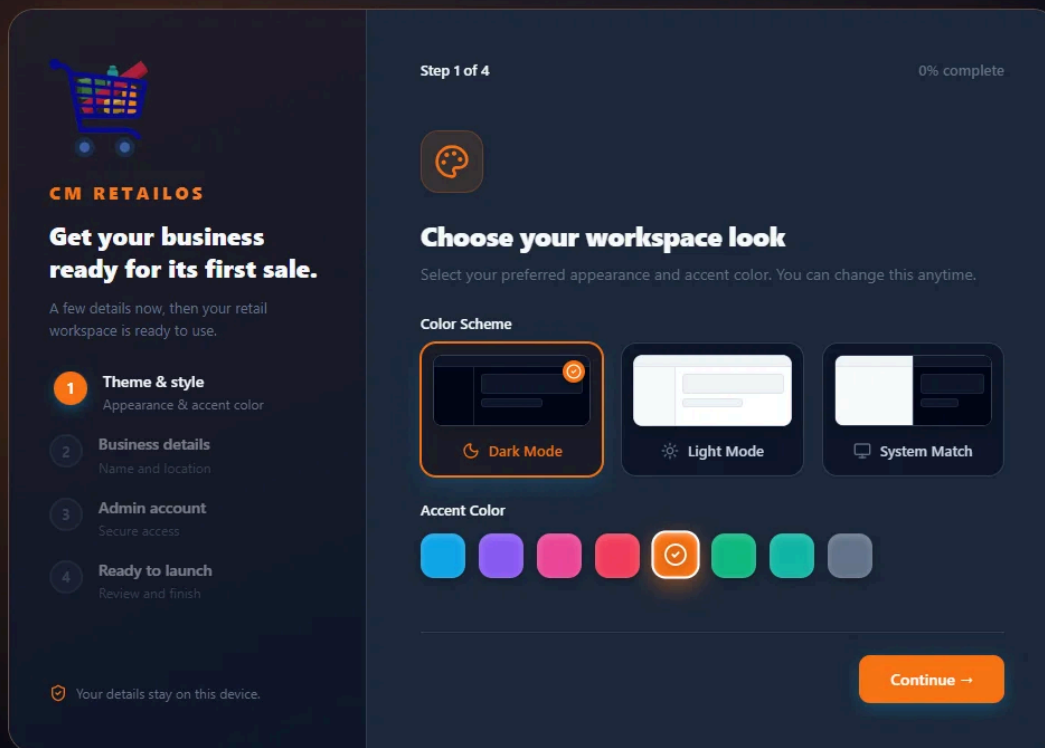


FIGURE 1.3

Step 1: Custom Amber Accent Selected

HOW TO READ & USE THIS SCREEN

Step 1 with a custom amber/orange accent selected over Dark Mode. The accent color is the single hue CM RetailOS uses for primary buttons, active sidebar items, links and chart highlights everywhere in the app.

- 1 Click several swatches and watch the previewed controls change color.
- 2 Choose an accent that clearly contrasts your scheme so actionable items stand out.
- 3 Continue when you are happy with the combination.

 **Pro tip:** Match the accent to your shop signage or logo color for a branded, professional feel on every screen.

The screenshot shows a mobile app interface for 'CM RETAILOS'. On the left, a sidebar lists four steps: 1. Theme & style (Appearance & accent color), 2. Business details (Name and location), 3. Admin account (Secure access), and 4. Ready to launch (Review and finish). Step 2 is currently active. Below the sidebar, a note states 'Your details stay on this device.' The main area is titled 'Step 2 of 4' with a progress bar at 33% complete. It features a storefront icon and the heading 'Tell us about your business', followed by the subtext 'We'll use this to personalise your workspace.' The form contains four input fields: 'Country' with a dropdown menu labeled 'Choose a country', 'Business or shop name' with a placeholder 'e.g. My Shop', 'Phone number' with a phone icon, and 'Physical address' with a location pin icon. At the bottom, there are 'Back' and 'Continue' buttons.

CM RETAILOS

Get your business ready for its first sale.

A few details now, then your retail workspace is ready to use.

- ✓ **Theme & style**
Appearance & accent color
- 2 Business details**
Name and location
- 3 **Admin account**
Secure access
- 4 **Ready to launch**
Review and finish

🛡️ Your details stay on this device.

Step 2 of 4 33% complete

Tell us about your business
We'll use this to personalise your workspace.

Country * **Business or shop name ***

📍 Choose a country 🏠 e.g. My Shop

Phone number **Physical address**

☎️ 📍

← Back Continue →

FIGURE 1.4

Step 2: Business Profile Form (Empty)

HOW TO READ & USE THIS SCREEN

Step 2 of 4 — “Tell us about your business.” The empty form waits for four required inputs: Country, Business or shop name, Phone number and Physical address. These details personalise the workspace and are printed on receipts and reports.

- 1 Choose your country first — it automatically sets your currency (Uganda → Uganda Shilling USh/UGX) and phone format.
- 2 Type the business name exactly as customers should see it on receipts.
- 3 Enter a phone number you check daily, plus the shop's physical address.
- 4 Press Continue once every marked field is filled.

⚠️ Watch out: Receipts, reports and the dashboard all reuse this information — a misspelled shop name will print on every single receipt until you fix it in Settings.



CM RETAILOS

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A few details now, then your retail workspace is ready to use.

- ✓ **Theme & style**
Appearance & accent color
- 2 Business details**
Name and location
- 3 **Admin account**
Secure access
- 4 **Ready to launch**
Review and finish

🔒 Your details stay on this device.

Step 2 of 4

33% complete



Tell us about your business

We'll use this to personalise your workspace.

Country *

Uganda

Business or shop name *

Real Supermarket

Uganda Shilling • USh UGX

Mobile Money Phone Number *

+256 759604799

Physical address

Nansana, Kyebando

✓ Valid number for Uganda (+256)

← Back

Continue →

FIGURE 1.5

Step 2: Completed Profile for "Real Supermarket"

📖 HOW TO READ & USE THIS SCREEN

A completed Step 2 for "Real Supermarket": country Uganda, currency Uganda shilling (USh UGX), Mobile Money phone +256 759604799 with the green "Valid number for Uganda (+256)" confirmation, and address Nansana, Kyebando. The progress bar now reads 33% complete.

- 1 Watch the phone field validate as you type — the helper line confirms the country format.
- 2 The Mobile Money number doubles as your licence-payment contact, so keep it active.
- 3 Continue unlocks only when all required fields pass validation.

CM RETAILOS

Get your business ready for its first sale.

A few details now, then your retail workspace is ready to use.

- ✓ **Theme & style**
Appearance & accent color
- ✓ **Business details**
Name and location
- 3 Admin account**
Secure access
- 4 **Ready to launch**
Review and finish

☒ Your details stay on this device.

Step 3 of 4 67% complete

Create your administrator account

This account has **full** access to your business workspace.

Your full name *

Username *

You will use this to sign in.

Create a password *

Confirm password *

Use at least 6 characters. Repeat your password.

[← Back](#) [Continue →](#)

FIGURE 1.6

Step 3: Administrator Account Setup (Empty)

HOW TO READ & USE THIS SCREEN

Step 3 of 4 — “Create your administrator account. This account has full access to your business.” You supply Your full name, a Username, a Password and Repeat password. This master account is the only one that can open Settings, see profit analytics and manage staff.

- 1 Enter the owner's full name — it appears in the profile menu and audit logs.
- 2 Choose a short username without spaces (for example `muheki`).
- 3 Type a strong password and repeat it exactly.
- 4 Never share this account with cashiers — create separate cashier accounts later instead.

⚠ Watch out: Whoever holds the administrator password can view every profit figure, change prices and manage staff. Treat it like your safe key.

CM RETAILOS

Get your business ready for its first sale.

A few details now, then your retail workspace is ready to use.

- Theme & style
Appearance & accent color
- Business details
Name and location
- Admin account**
Secure access
- Ready to launch
Review and finish

Your details stay on this device.

Step 3 of 4 67% complete

Create your administrator account

This account has **full** access to your business workspace.

Your full name * Username *

Christopher Muheki muheki

You will use this to sign in.

Create a password * Confirm password *

At least 6 characters. ✓ Passwords match

← Back Continue →

FIGURE 1.7

Step 3: Master Admin Credentials Validated

HOW TO READ & USE THIS SCREEN

Step 3 filled in for Christopher Muheki (username **muheki**) with the live “Passwords match” confirmation showing — that confirmation is what enables the Continue button. The wizard re-checks both password boxes on every keystroke.

- 1 If “Passwords match” does not appear, re-type both fields slowly — one mismatched character blocks you.
- 2 Once confirmed, press Continue to reach the final review step.

Pro tip: Use a passphrase of three random words plus a number — long, memorable, and hard to guess at a shared counter.



CM RETAILOS

Get your business ready for its first sale.

A few details now, then your retail workspace is ready to use.

- ✓ Theme & style
Appearance & accent color
- ✓ Business details
Name and location
- ✓ Admin account
Secure access
- 4 Ready to launch
Review and finish

🔒 Your details stay on this device.

Step 4 of 4

100% complete



You're all set!

Review your details before launching.

Shop Name	Real Supermarket
Country	Uganda
Phone	759604799
Address	Nansana, Kyebando
Admin Name	Christopher Muheki
Username	muheki

← Back

Launch →

FIGURE 1.8

Setup Wizard — Step 4: "You're all set!" Verification Summary (Shop Name, Country, Address, Admin Name)

📖 HOW TO READ & USE THIS SCREEN

Step 4 of 4 — "You're all set! Review your details before launching." The summary lists Shop Name (Real Supermarket), Country (Uganda), Phone, Address (Nansana, Kyebando), Admin Name and username at 100% complete. Nothing is written to the workspace until you approve this page.

- 1 Read every row carefully — this is your last chance to correct anything using Back.
- 2 Press the launch/finish button to create the offline database and hand over to the activation screen.
- 3 Write the admin credentials down and store them safely.

Hardware-Bound Activation & Air-Gapped Licensing

CM RetailOS uses a secure, cryptographic licensing system tied directly to your computer's motherboard and CPU hardware ID. Works 100% offline with zero cloud dependency.

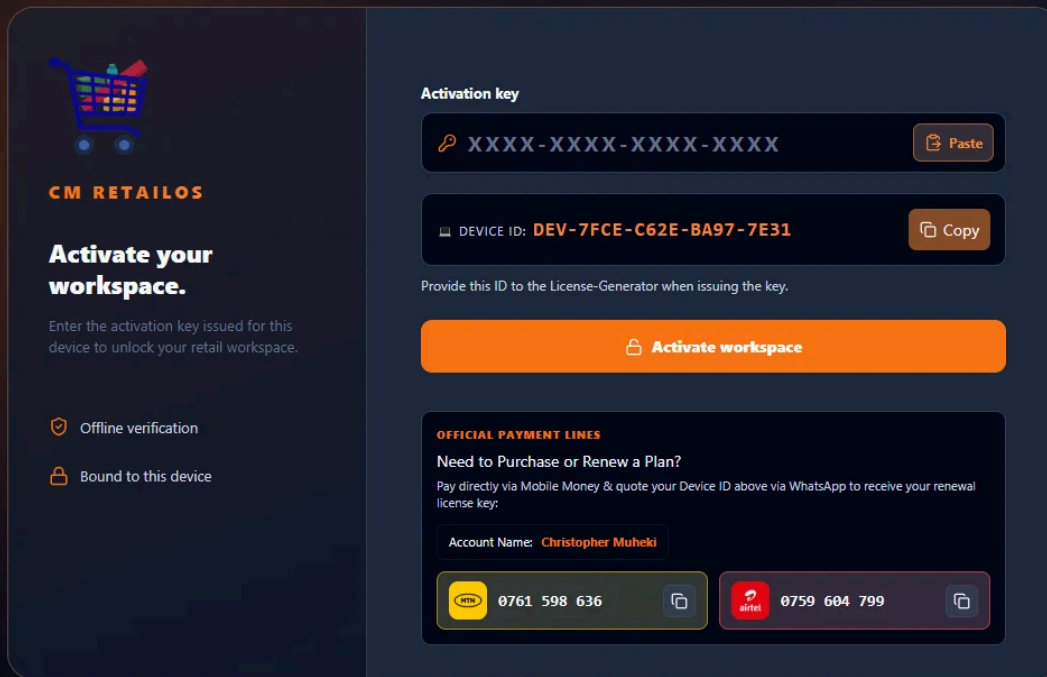


FIGURE 2.1

Workspace Activation — Unique Hardware Device ID Extraction (`DEV-7FCE-C62E-8A97-7E31`) & Mobile Money Payment Channels

HOW TO READ & USE THIS SCREEN

The Workspace Activation screen. CM RetailOS has already extracted this computer's unique hardware fingerprint — Device ID `DEV-7FCE-C62E-8A97-7E31` — and lists the OFFICIAL PAYMENT LINES (MTN & Airtel Mobile Money, account name Christopher Muheki). The two badges explain the design: *Offline verification* and *Bound to this device*.

- 1 Click the copy button next to the Device ID to copy it exactly.
- 2 Pay the licence fee (**5,000 UGX per month**) via MTN MoMo or Airtel Money to the official lines shown.
- 3 WhatsApp your Device ID plus the payment reference to the vendor.
- 4 When your activation key arrives, paste it into the `XXXX-XXXX-XXXX-XXXX` field with the Paste button and activate.

⚠ Watch out: Licence keys are cryptographically bound to the Device ID that requested them. A key bought for another computer will be rejected here — always copy the ID from this exact screen.

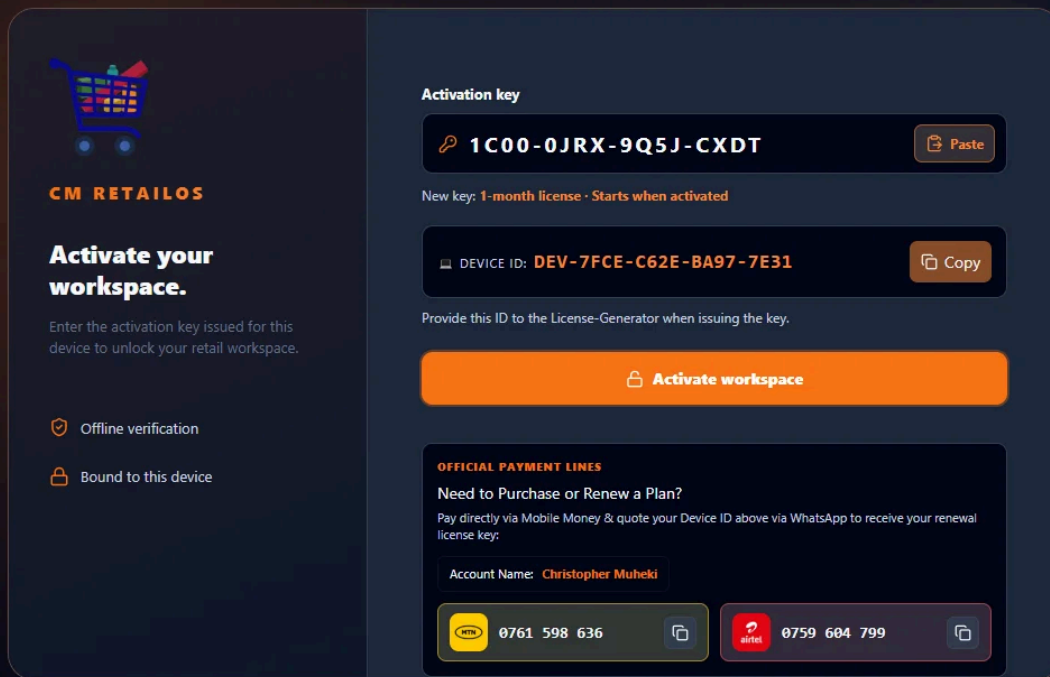


FIGURE 2.2 Activation Key Validated (`1C00-0JRX-9Q5J-CXDT`)

HOW TO READ & USE THIS SCREEN

The same screen after pasting a purchased key, `1C00-0JRX-9Q5J-CXDT`. The notice “New key: 1-month license – Starts when activated” tells you the countdown begins only when activation succeeds, not when you buy the key.

- 1 Paste the 16-character key exactly as received — four groups of four characters.
- 2 Press Activate; verification happens locally on the machine, no internet needed.
- 3 On success the workspace unlocks and the 30-day period starts.

 **Pro tip:** Activate on the morning of a busy trading day so your first licensed day covers a full day of sales.

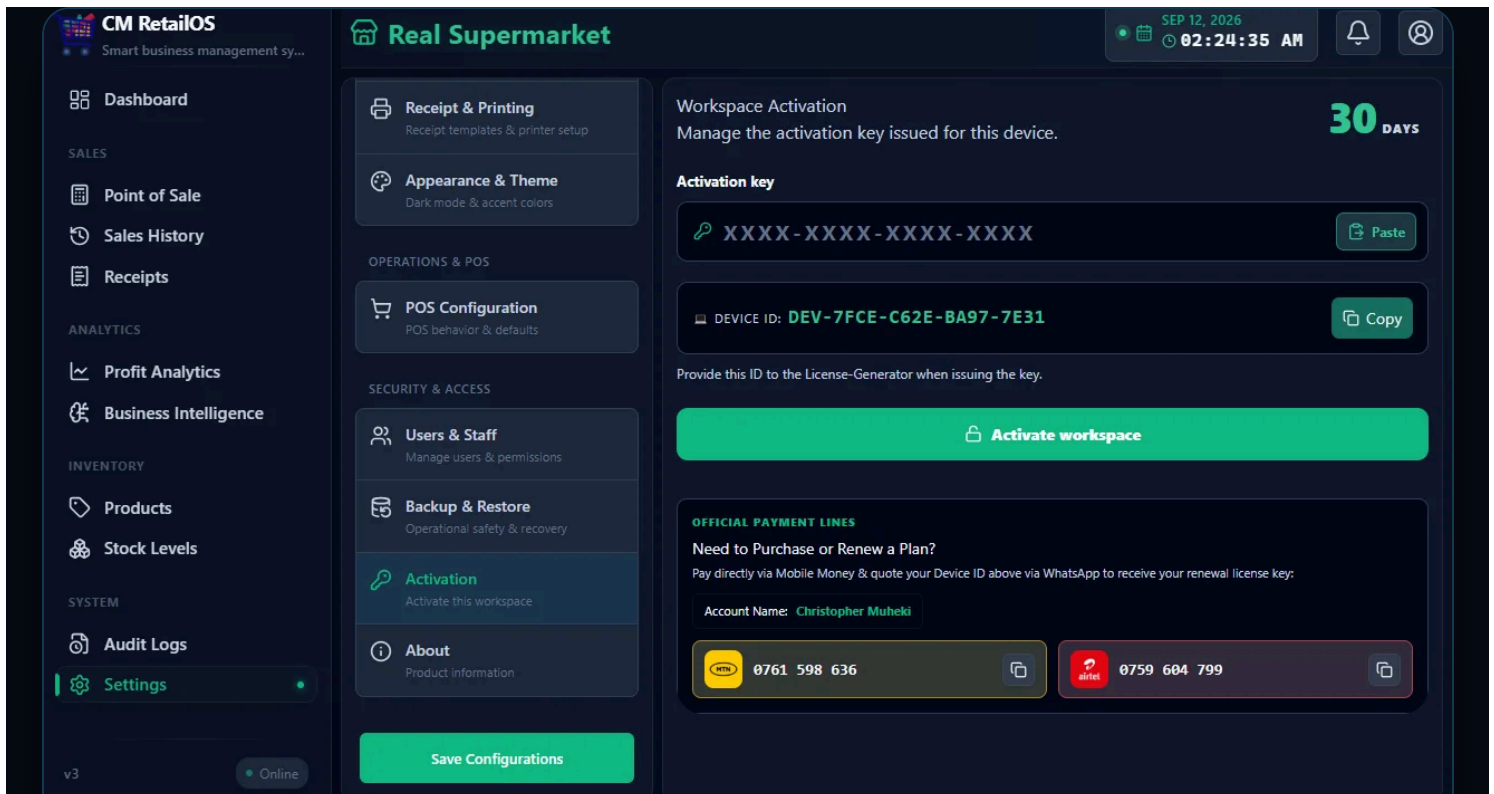


FIGURE 2.3

Settings > Activation: Active Workspace (30 Days Remaining)

HOW TO READ & USE THIS SCREEN

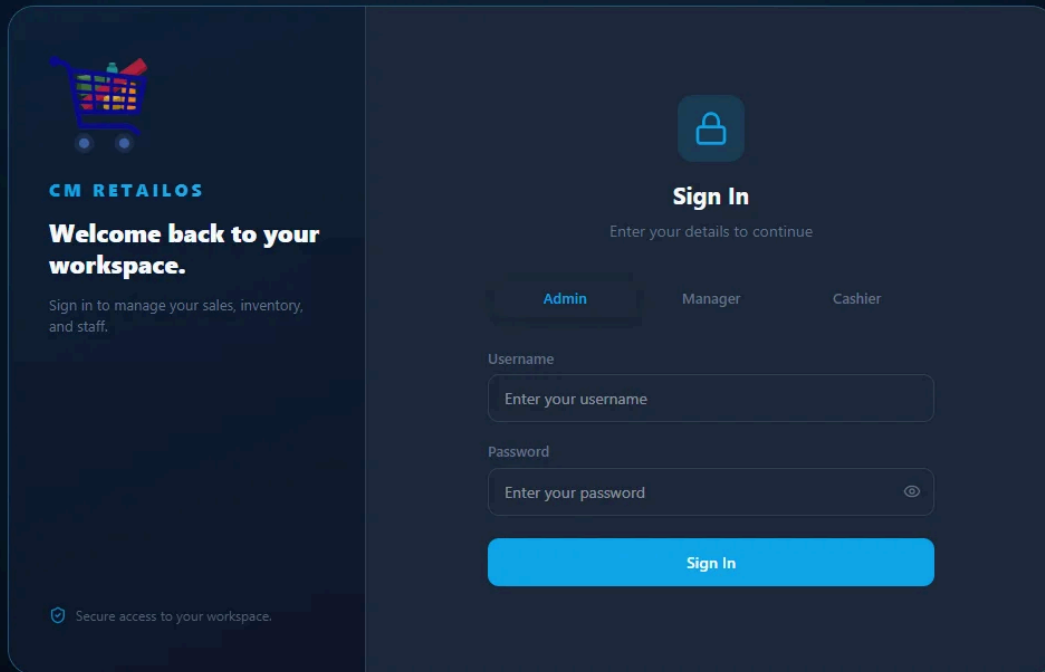
Settings → Activation, the in-app home of your licence. It repeats your Device ID, shows the activation key field, the remaining-days gauge (about 30 days on a fresh 1-month licence at 5,000 UGX/month, grace period noted) and the same official Mobile Money payment lines for renewals.

- 1 Open Settings → Activation once a week and glance at the days remaining.
- 2 Renew 2–3 days before expiry using the same Device ID flow so trading is never interrupted.
- 3 After pasting a renewal key the countdown resets immediately.

⚠ Watch out: When the licence lapses the workspace locks back to the activation screen — sales cannot be rung up until a valid key is entered, so never leave renewal to the last minute.

Workspace Authentication & Multi-Role Sign-In

Role-Based Access Control (RBAC) ensures cashiers are locked strictly to checkout counters while administrators access financial profit ledgers.



The image shows a mobile app sign-in screen for 'CM RETAILOS'. The left panel features a shopping cart icon, the company name, a welcome message, a description of the app's purpose, and a security badge. The right panel is titled 'Sign In' with a lock icon and a subtitle 'Enter your details to continue'. It includes three role tabs: 'Admin' (highlighted in red), 'Manager', and 'Cashier'. Below these are input fields for 'Username' and 'Password', each with a placeholder text. A red 'Sign In' button is at the bottom of the right panel.

FIGURE 3.1 Sign-In Gateway — Fast Role Switcher Tabs (Admin / Manager / Cashier)

HOW TO READ & USE THIS SCREEN

The sign-in gateway — “Welcome back to your workspace.” One window serves the whole team: role tabs (Admin / Manager / Cashier) sit above the Username and Password fields. The role you pick decides which menus and figures you will see after signing in.

- 1 Tap the tab that matches your role.
- 2 Enter your username and password exactly as created by the administrator.
- 3 Press Sign In — admins land on the full dashboard, cashiers on the locked terminal.

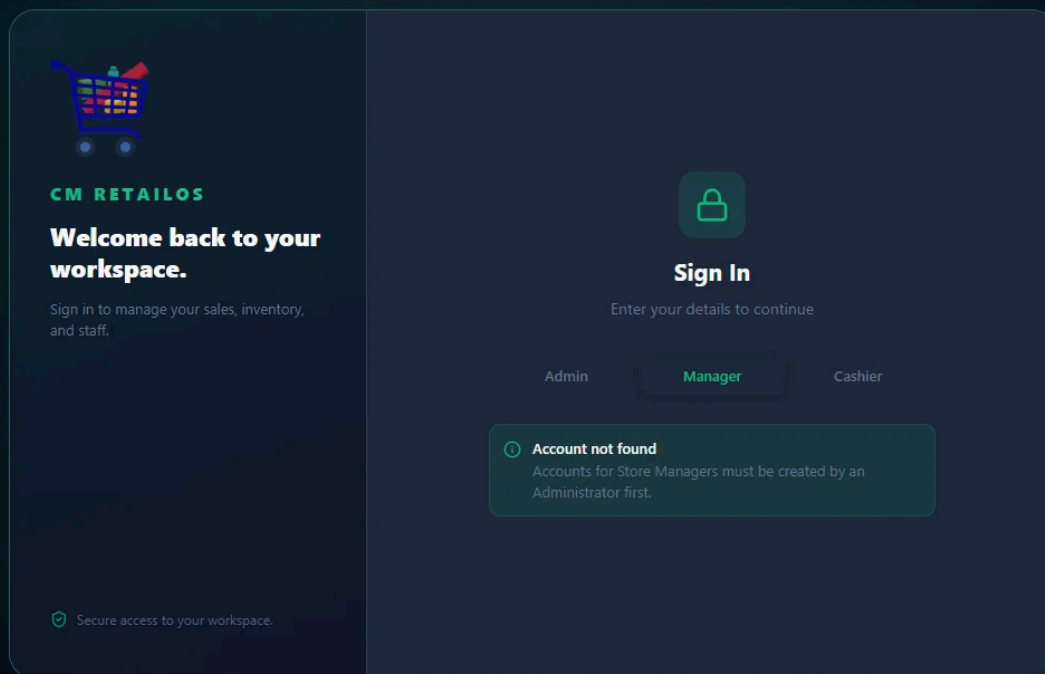


FIGURE 3.2

Role Guard — "Account not found" Notice on Unprovisioned Manager Tab

HOW TO READ & USE THIS SCREEN

Role provisioning guard in action: when clicking the Manager tab before any Store Manager account has been created, CM RetailOS displays an informative guidance notice: *"Account not found — Accounts for Store Managers must be created by an Administrator first."* The sign-in form is safely withheld until an administrator provisions the account.

- 1 Sign in under the **Admin** tab first using your primary administrator credentials.
- 2 Navigate to **Settings > Users & Staff** to add and configure a Store Manager profile.
- 3 Once created, returning to the Manager tab will unlock the Manager username and password sign-in fields.

 **Pro tip:** Store Managers have elevated operational privileges (sales history, stock adjustments, and staff oversight), but system-wide master settings and root database tools remain strictly reserved for the Administrator.

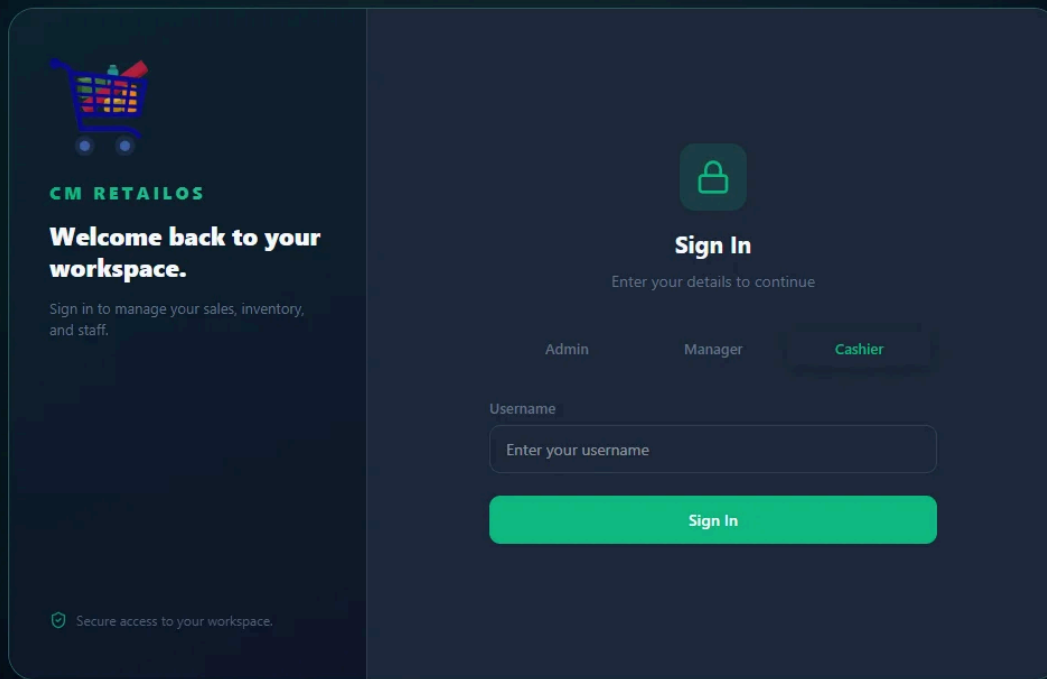


FIGURE 3.3

Cashier Sign-In — Fast Username-Only Login (No Password or PIN Required)

HOW TO READ & USE THIS SCREEN

The dedicated Cashier sign-in mode branded “Secure access to your workspace”. Built for maximum counter efficiency: cashiers only need to enter their assigned Username to access their terminal. There are no passwords or PINs required for cashiers, enabling rapid lane handovers while keeping sensitive business data completely protected.

- 1 Click the **Cashier** role tab.
- 2 Type the cashier's assigned **Username** into the single input field.
- 3 Press **Sign In** — the checkout terminal opens instantly, while cost prices, profit ledgers, and administrative settings remain strictly locked out.
- 4 At the end of a shift, click the profile icon and select Switch Account so the next cashier can log in with their own username.

 **Pro tip:** Because cashiers have zero access to profit reports or system configurations, username-only login provides fast, frictionless shift handovers without compromising store security.

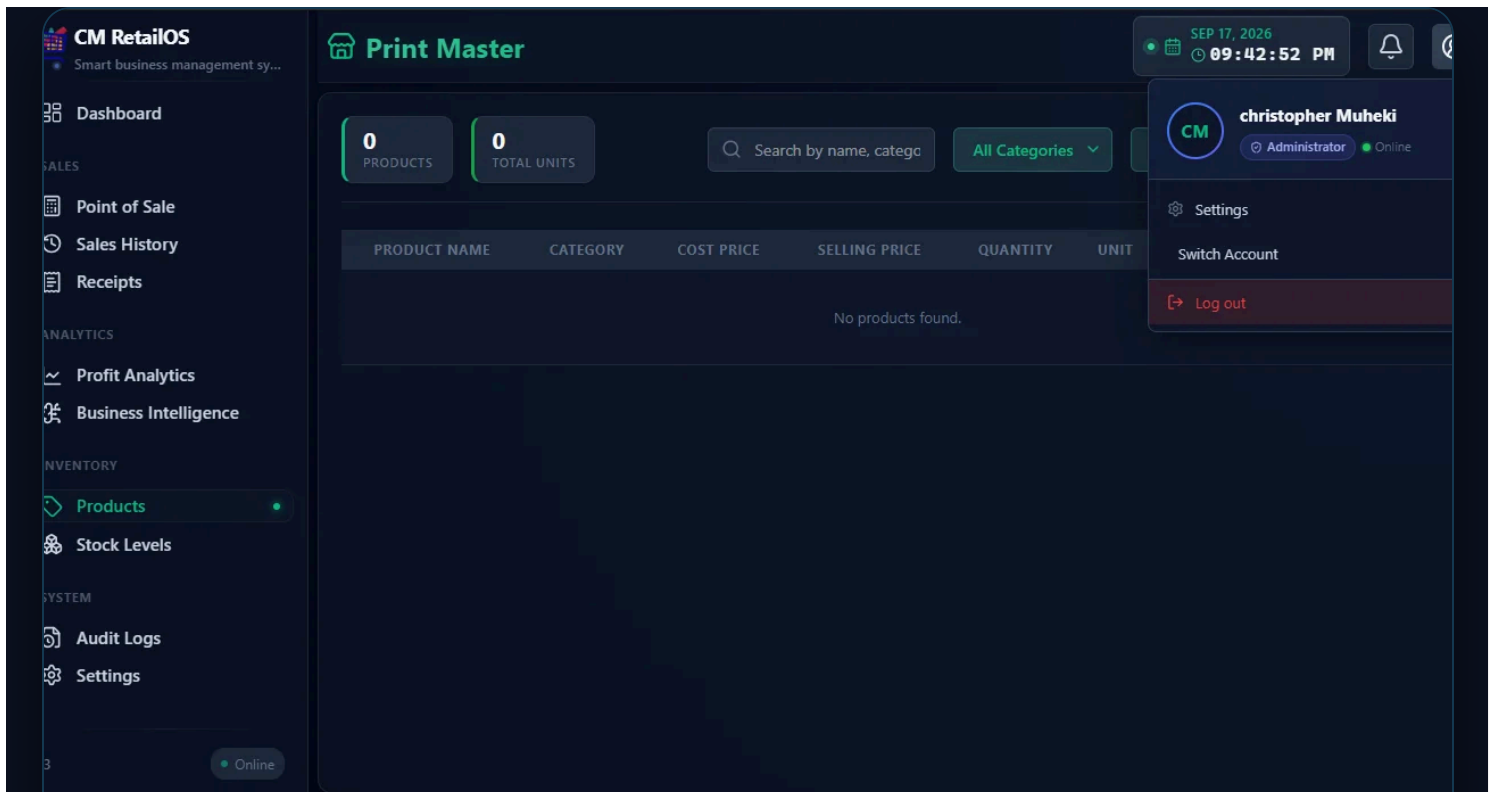


FIGURE 3.4

Header Profile Menu (Active User: Christopher Muheki, Admin)

HOW TO READ & USE THIS SCREEN

The header profile menu opened for Christopher Muheki (Administrator). It confirms who is currently signed in and their role, and offers Settings plus Switch Account without closing the app.

- 1 Click your name at the top-right any time to verify the active user.
- 2 Use Switch Account to hand the counter to another staff member cleanly.
- 3 Admins can jump straight into Settings from this menu.

⚠ Watch out: Never walk away from an unlocked admin session — switch account or sign out before leaving the desk.

Executive Dashboard & Operational Telemetry

Monitor real-time sales revenue, estimated net profit, orders volume, inventory valuation, hourly traffic curves, and automated stockout alerts.



FIGURE 4.1 Dashboard Zero-State Baseline for "My Duuka"

HOW TO READ & USE THIS SCREEN

The executive dashboard on day one for "My Duuka": Today's Revenue USh 0, Today's Profit USh 0, Orders Today 0, Total Inventory Value USh 0, an empty 7am–11pm Sales Overview curve and an empty Top Selling Products list. This is the zero-state baseline before the first sale is rung up.

- 1 Read the four KPI cards as your daily health check.
- 2 Understand that every number is live — each completed POS sale updates revenue, profit and orders instantly.
- 3 Once data exists, use the Daily / Weekly / Monthly tabs to change the chart window.

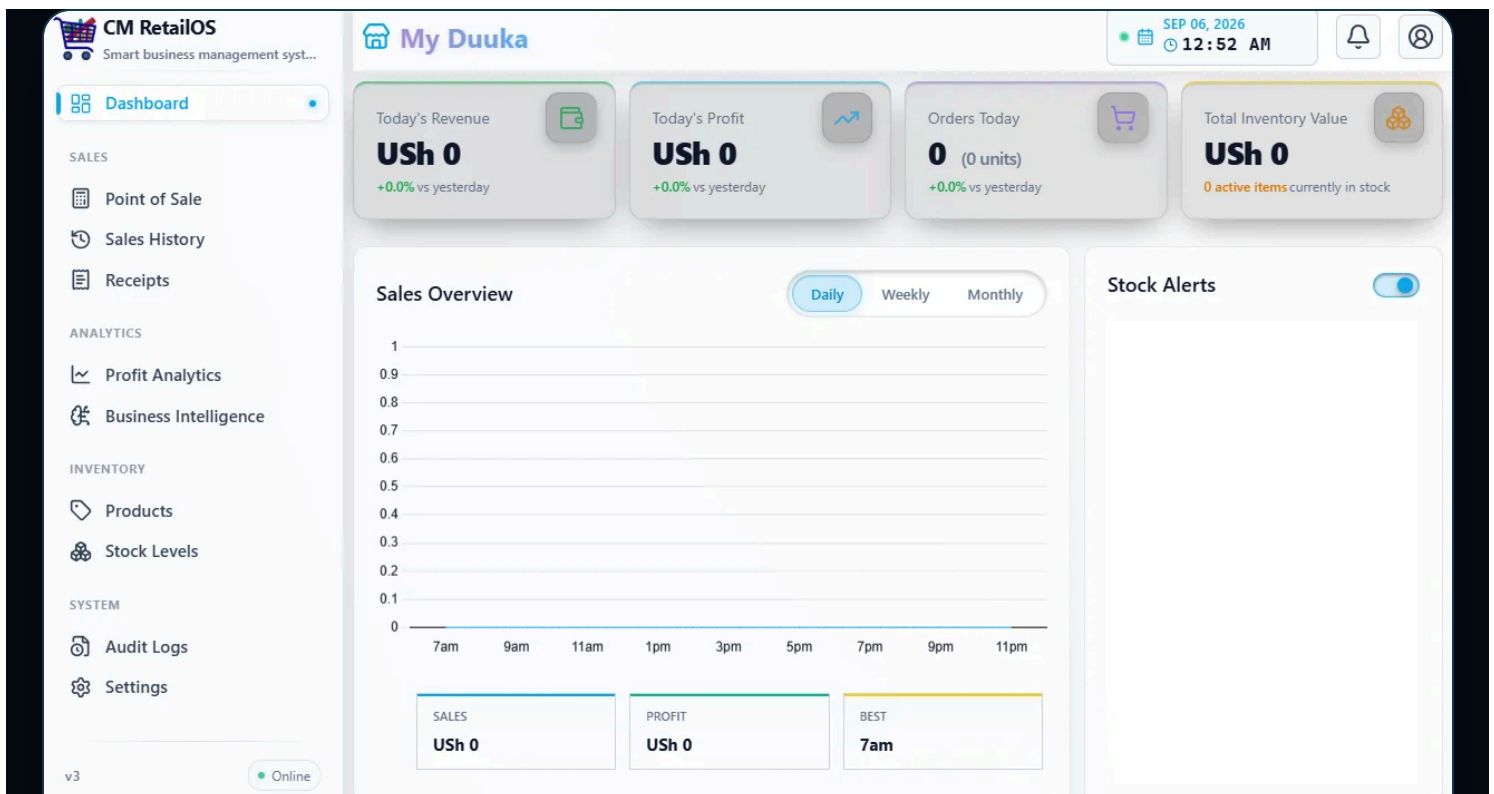


FIGURE 4.2 Left Sidebar Navigation (Sales, Analytics, Inventory, System)

HOW TO READ & USE THIS SCREEN

The dashboard annotated by its left sidebar — the map of the whole application: SALES (Point of Sale, Sales History, Receipts), ANALYTICS (Profit Analytics, Business Intelligence), INVENTORY (Products, Stock Levels) and SYSTEM (Audit Logs, Settings). The header carries the live date and clock (SEP 06, 2026 12:52 am) plus an Online status badge.

- 1 Memorise the four groups — every task in this manual lives under one of them.
- 2 Click any item to switch modules instantly; the active item is highlighted.
- 3 Use the header clock for shift awareness and the badge to confirm workspace status.

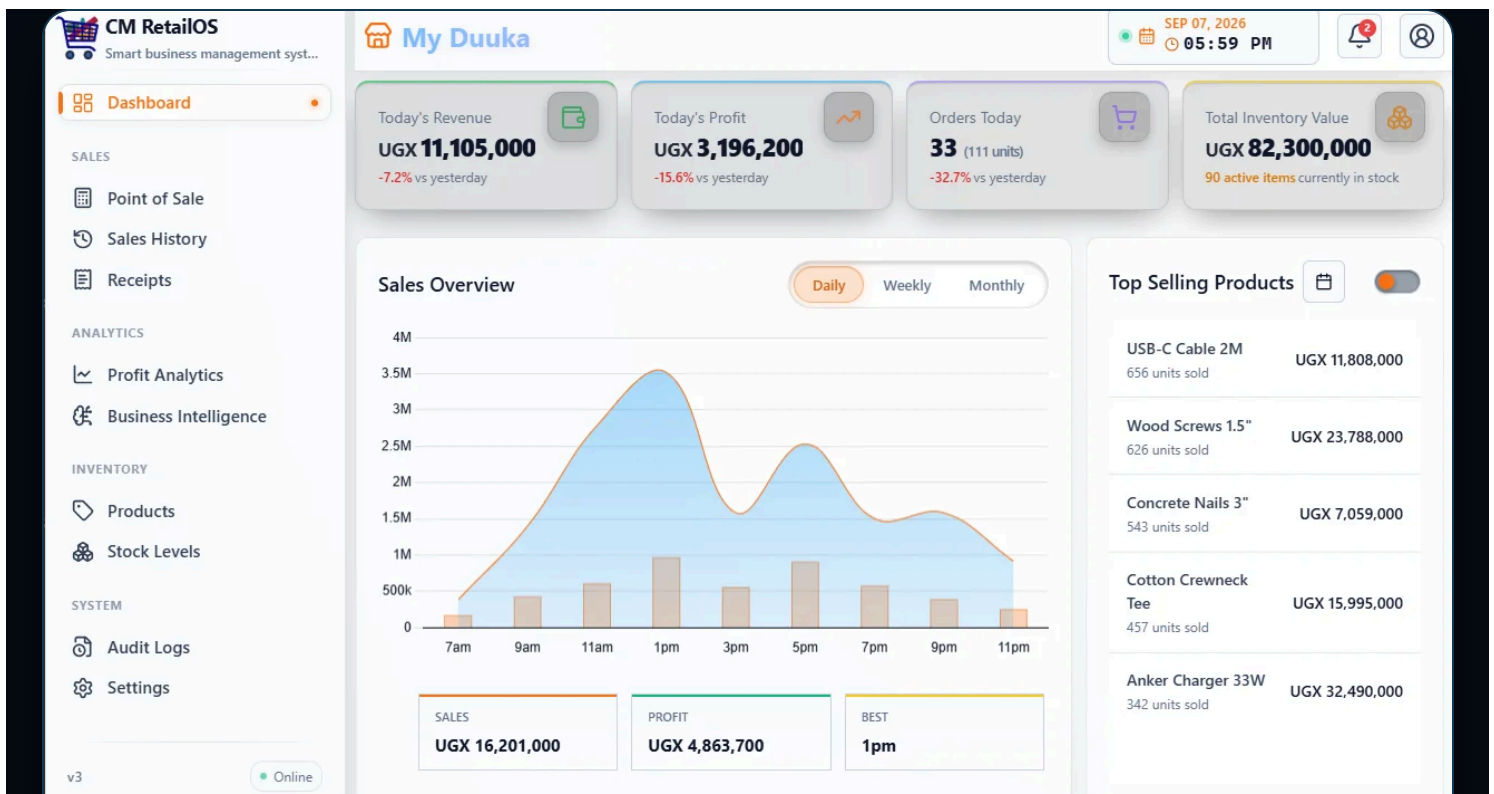


FIGURE 4.3

Live Dashboard Telemetry (Light Mode) — UGX 11.1M Daily Revenue, UGX 3.19M Profit, 33 Orders & Top SKUs Leaderboard

HOW TO READ & USE THIS SCREEN

A live trading day in Light Mode: UGX 11,105,000 revenue, UGX 3,196,200 profit, 33 orders — each with “vs yesterday” percentages — a hoverable hourly curve (the tooltip reads Sales UGX 16,201,000 / Profit UGX 4,863,700 at 1pm), Total Inventory Value UGX 82,300,000 across 90 active items, and the Top Selling Products leaderboard led by USB-C Cable 2M (656 units).

- 1 Start each morning here: revenue tells you volume, profit tells you quality.
- 2 Hover any point of the curve to see exact hourly sales and profit.
- 3 Scroll the Top Selling Products list to decide what to restock first.
- 4 Flip Daily / Weekly / Monthly to spot trends beyond a single day.



FIGURE 4.4

Live Dashboard (Dark Mode) — Stock Alerts Panel & 7pm Peak Tooltip

HOW TO READ & USE THIS SCREEN

The same dashboard in Dark Mode at 6:01 PM, focused on the Stock Alerts panel: Quilted Shoulder Tote (low stock, 2 left), Aviator Sunglasses (out of stock, 0), Denim Jacket Vintage (low, 9) and Fleece Neck Scarf (out) — each with an Adjust shortcut — plus a tooltip marking the 7pm sales peak.

- 1 Open the Stock Alerts panel every morning and after big sales days.
- 2 Click Adjust beside any alert to jump straight to that product's stock editor.
- 3 Treat "out of stock" rows as lost revenue — restock or hide them from the POS.

Pro tip: The hourly tooltip (7pm peak here) tells you when to schedule your strongest cashier and when the cash float needs topping up.



FIGURE 4.5 Live Dashboard for "ug electronics" (USh 7.58M Revenue, USh 2.32M Profit)

HOW TO READ & USE THIS SCREEN

The dashboard running a second business, "ug electronics" (USh 7.58M revenue, USh 2.32M profit). Everything — header shop name, currency labels, KPIs and leaderboards — is re-branded per workspace from the details entered in the setup wizard.

- 1 Recognise your own store by the header name before trading each day.
- 2 Keep one licensed PC per shop: workspaces are bound to both hardware and business identity.
- 3 Use Settings → Business Information to correct branding details at any time.



FIGURE 4.6

System Notifications Center — Live Stockout Warnings, Inventory Drops & Best Seller Alerts

HOW TO READ & USE THIS SCREEN

The notifications centre (bell icon, top-right). It collects live operational events — stockout warnings, inventory drops and best-seller alerts — in one feed so you never have to hunt for them.

- 1 Click the bell whenever the badge shows unread items.
- 2 Work the feed top-down: stockouts first, then low stock, then informational alerts.
- 3 Use the linked actions to jump to the affected product or report.

Pro tip: Make the notification feed part of your closing routine — five minutes each evening prevents tomorrow's stockouts.

Inventory, SKU Management & Android Scanner Bridge

Manage unlimited products, track stock thresholds, calculate automatic profit margins, and transform any Android phone into a high-speed camera scanner.

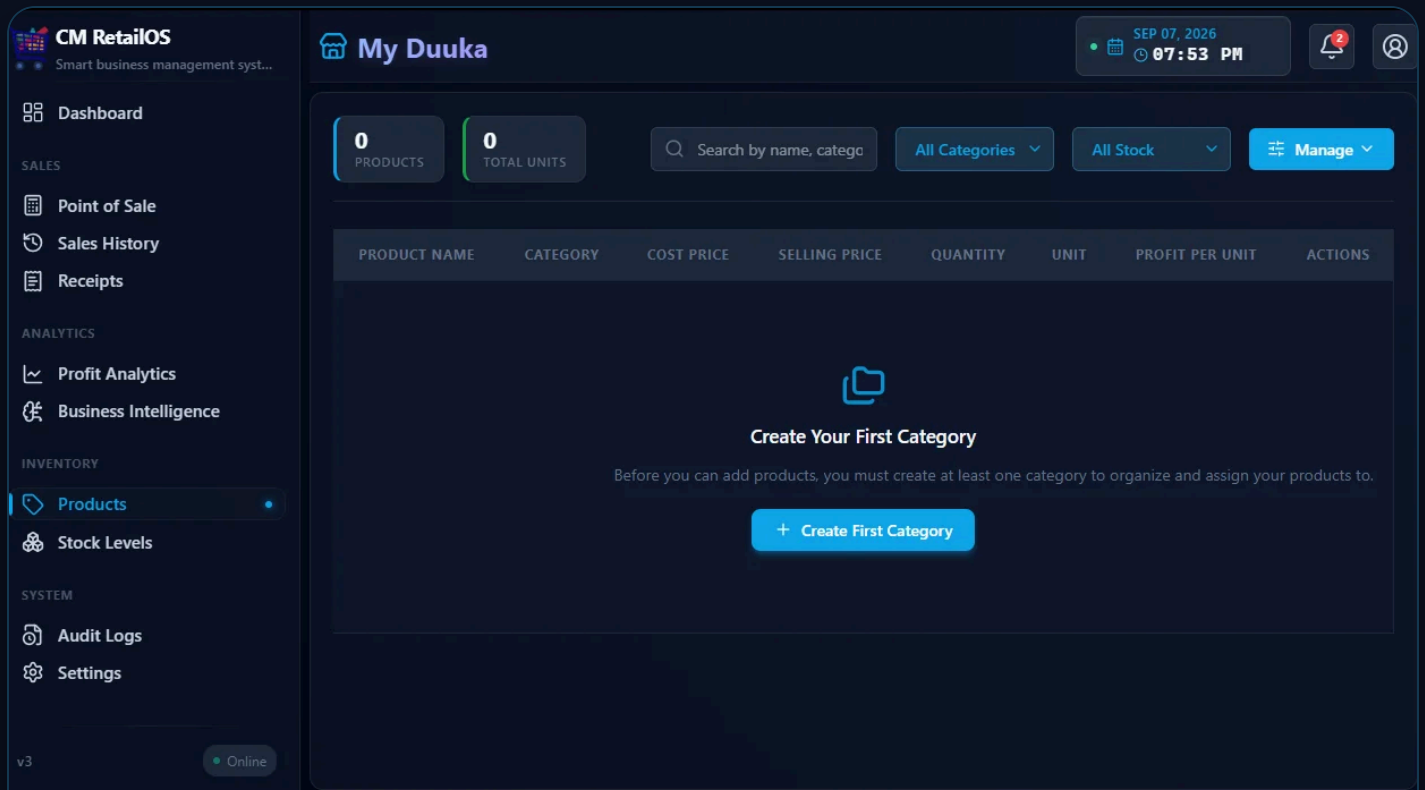


FIGURE 5.1 Category Initialization Prompt ("Create Your First Category")

HOW TO READ & USE THIS SCREEN

The Products page on a fresh install, blocked by the "Create Your First Category" prompt: "Before you can add products, you must create at least one category to organize and assign your products to." CM RetailOS insists on categories first because they power the POS tiles, the filters and the profit-by-category charts.

- 1 Click the prompt and create realistic shop aisles — e.g. Beverages, Bakery, Electronics, Fashion.
- 2 Keep the list short and stable; you can always add more later.
- 3 Once one category exists, the Add Product button unlocks.

FIGURE 5.2

Add New Product Modal — Auto SKU, EAN-13 Barcode Scan Field, Cost vs Selling Price with Live MARGIN, Min Stock & Unit Selector (pcs/kg/L/g/ml/pack/box)

HOW TO READ & USE THIS SCREEN

The Add New Product modal over the products table. Fields shown: Product Name (placeholder “eg. Fresh Orange Juice 1L”), an auto-generated SKU (**SKU-991046**), Category picker, Barcode “Scan or enter...” with EAN-13 format, Cost Price, Selling Price with an automatic MARGIN readout, Qty, Min Stock and the unit selector (pcs / kg / L / g / ml / pack / box).

- 1 Type the product name exactly as it should print on receipts.
- 2 Pick its category, then scan the barcode with your laser or Android scanner — or type the EAN-13 manually.
- 3 Enter cost and selling prices; the margin badge instantly shows your profit per unit.
- 4 Set opening Qty, a Min Stock threshold that will trigger Low Stock alerts, and the correct unit.
- 5 Save — the product is immediately sellable at the POS.

Pro tip: Set Min Stock to roughly one week of expected sales so the reorder alert arrives while the supplier can still deliver in time.

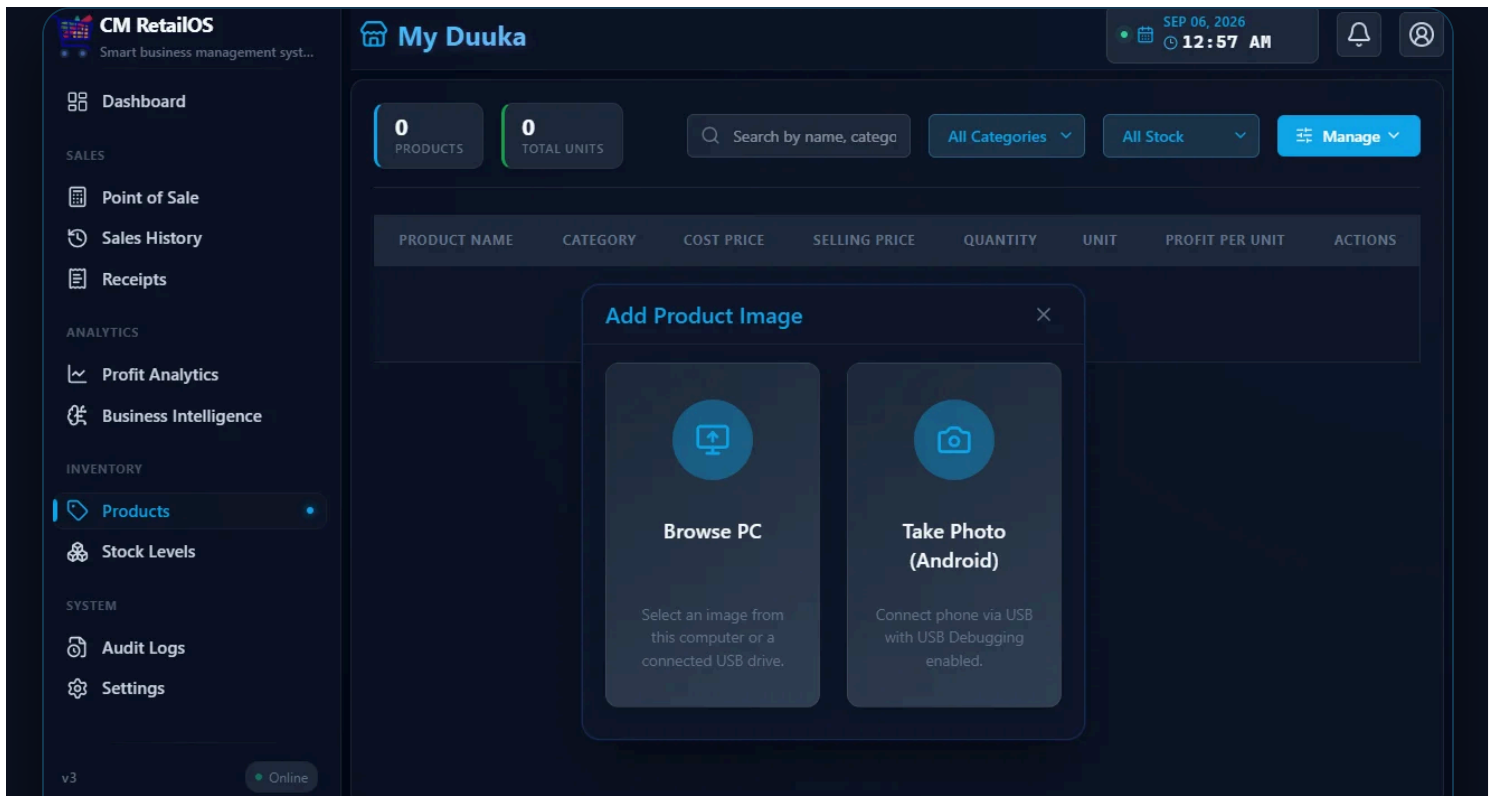


FIGURE 5.3

Add Product Image — "Browse PC" File Picker vs "Take Photo (Android)" USB Camera Scanner Bridge

HOW TO READ & USE THIS SCREEN

The Add Product Image dialog with its two sources: "Browse PC — Select an image from this computer or a connected USB drive" and "Take Photo (Android) — Connect phone via USB with USB Debugging enabled." The second option turns any Android phone into the shop's camera and barcode scanner.

- 1 For Browse PC: pick any JPG/PNG from the computer or a USB stick.
- 2 For Android: enable Developer Options → USB debugging on the phone, connect the USB cable, then choose Take Photo (Android).
- 3 Frame the product or its barcode and capture — the image attaches to the SKU and shows on POS tiles.

 **Pro tip:** The Android bridge means you never need to buy a separate barcode laser — a used smartphone plus a USB cable covers scanning and product photos.

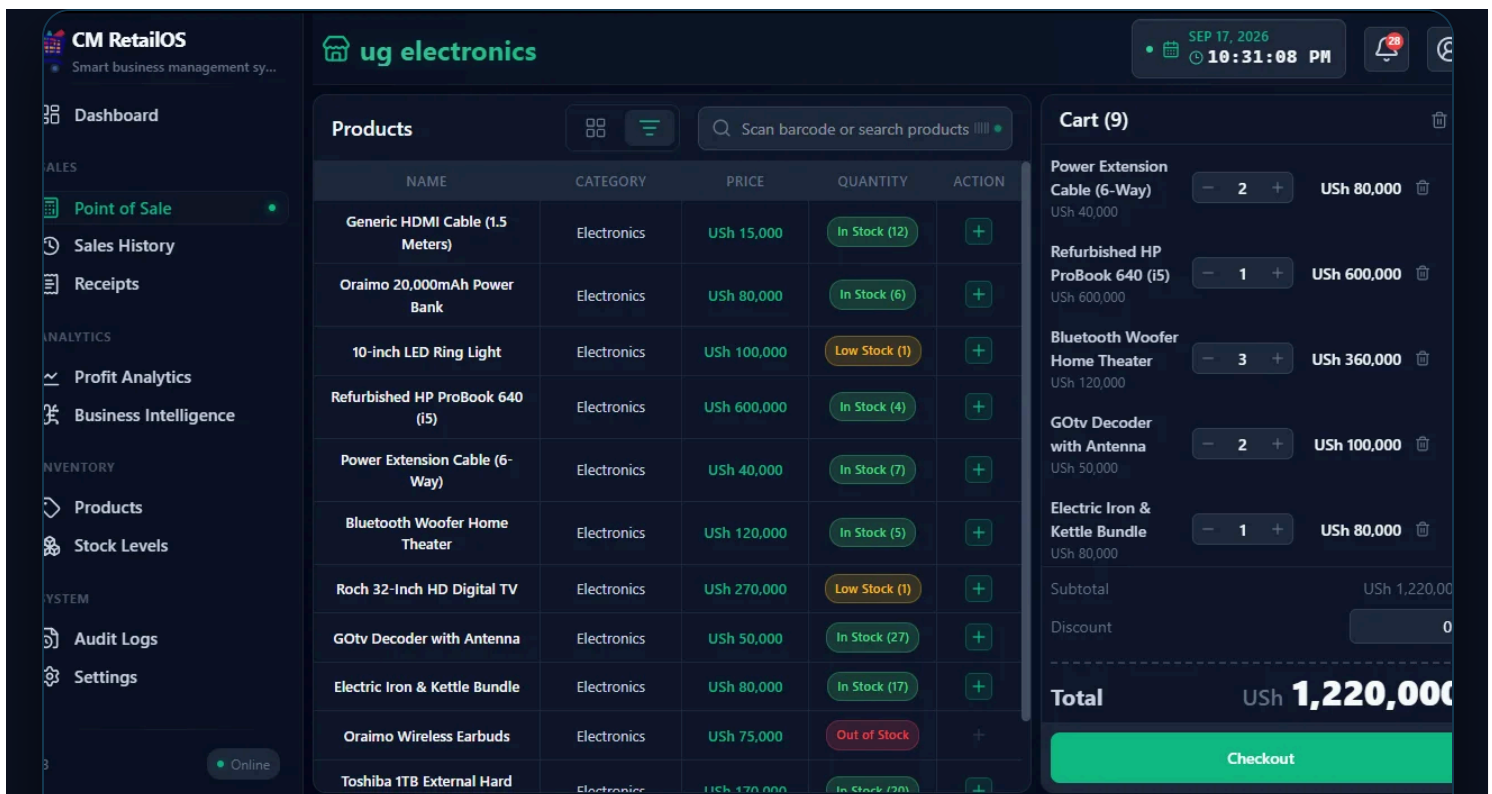


FIGURE 5.4

Populated Product Rows with Real-Time Stock Badges (In Stock / Low Stock) & Per-Item Actions

HOW TO READ & USE THIS SCREEN

A populated product list (fashion store example) showing the columns you manage daily — Name, Category and Price, with Quantity rendered as live stock badges such as In Stock and Low Stock — and the search box plus category/stock filters above the table.

- 1 Use the search box or the All Categories / All Stock filters to slice the catalog.
- 2 Read the badges at a glance: green In Stock, amber Low Stock, red Out of Stock.
- 3 Open a row's actions to edit price, restock quantity or attach a barcode.

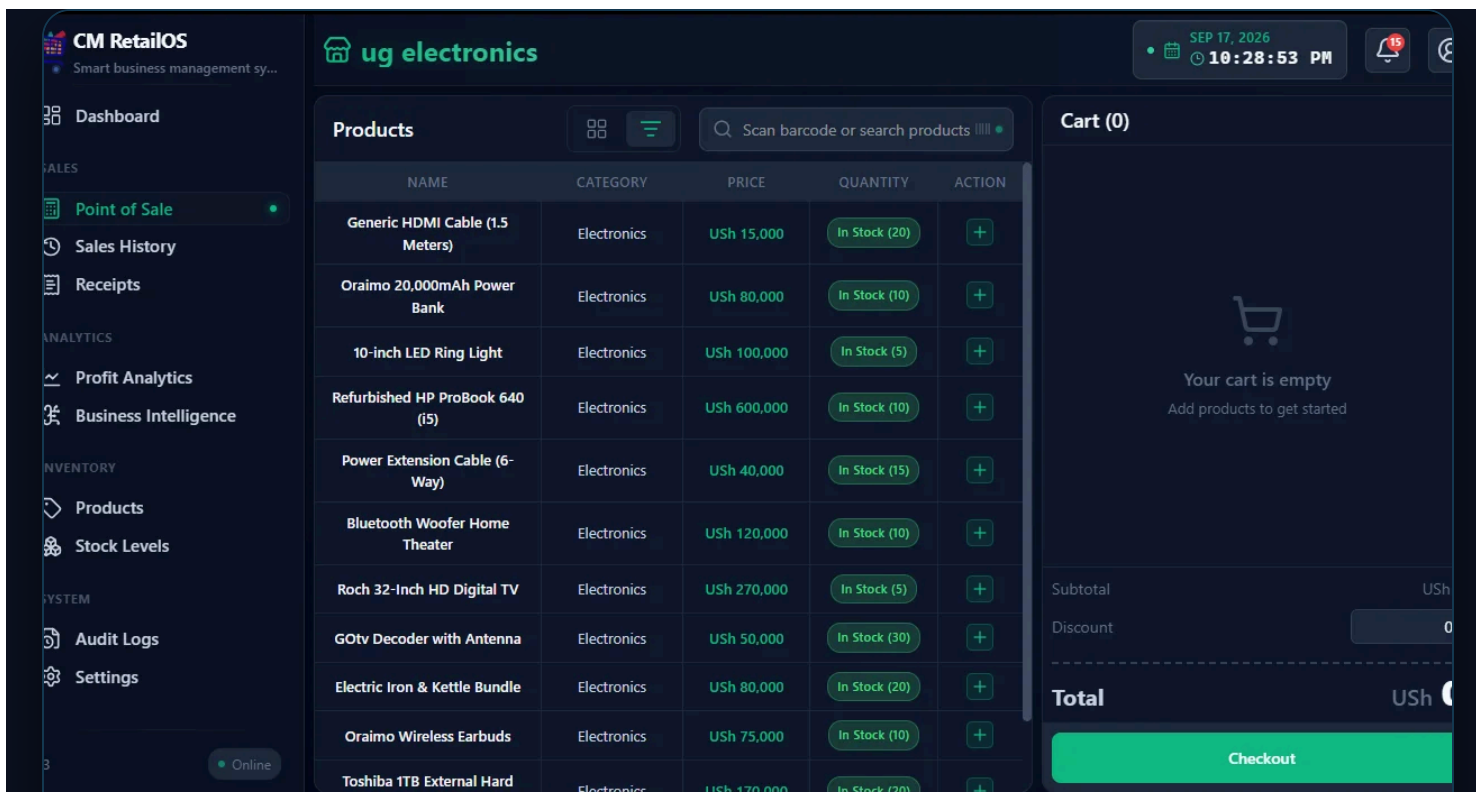


FIGURE 5.5

Selling Lane Table with Live Stock Counts per SKU — In Stock (20), Low Stock & Out of Stock Badges Update with Every Sale

HOW TO READ & USE THIS SCREEN

The selling lane in table mode for “ug electronics”, where every SKU row carries its live stock count — In Stock (20), In Stock (10), In Stock (5) ... — beside the price and a one-tap add button, with “Scan barcode or search products” at the top.

- 1 Watch stock counts fall in real time as sales post — this is your live shelf.
- 2 When a count approaches zero, reorder or transfer stock before it hits Out of Stock.
- 3 Teach cashiers that the scan box accepts a laser scanner, the Android bridge, or typed SKUs.

Point of Sale (POS) & Cashier Checkout Lanes

Lightning-fast cashier terminal supporting barcode laser scanning, visual touch tile catalog, cash checkout, and instant thermal receipt printing.

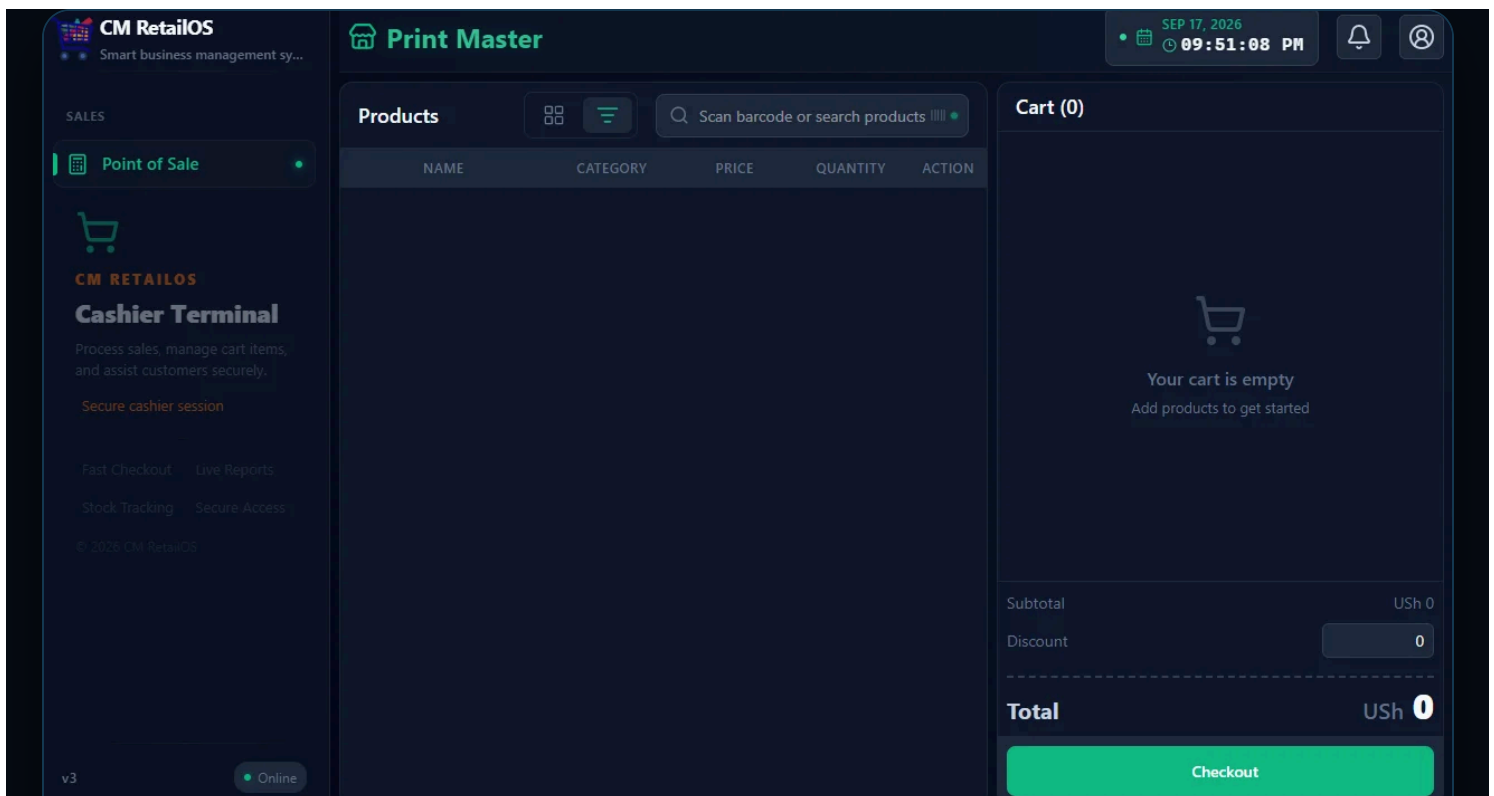


FIGURE 6.1

Dedicated Cashier Terminal Session (Locked Mode — Profit & Cost Prices Hidden)

HOW TO READ & USE THIS SCREEN

The dedicated Cashier Terminal in secure session mode: product lane on the left (Name, Category, Price, Quantity, Action), an empty Cart (0) on the right with Subtotal / Discount / Total at USh 0, and a Checkout button. Cost prices and profit figures are deliberately absent — cashiers see selling prices only.

- 1 Find items via the search/scan box or by clicking a row's + button.
- 2 Each add appears in the cart with quantity and line total.
- 3 Use the trash icon (or the supervisor-approved flow) to remove mis-scans.
- 4 When the customer is ready, press Checkout to record the cash sale.

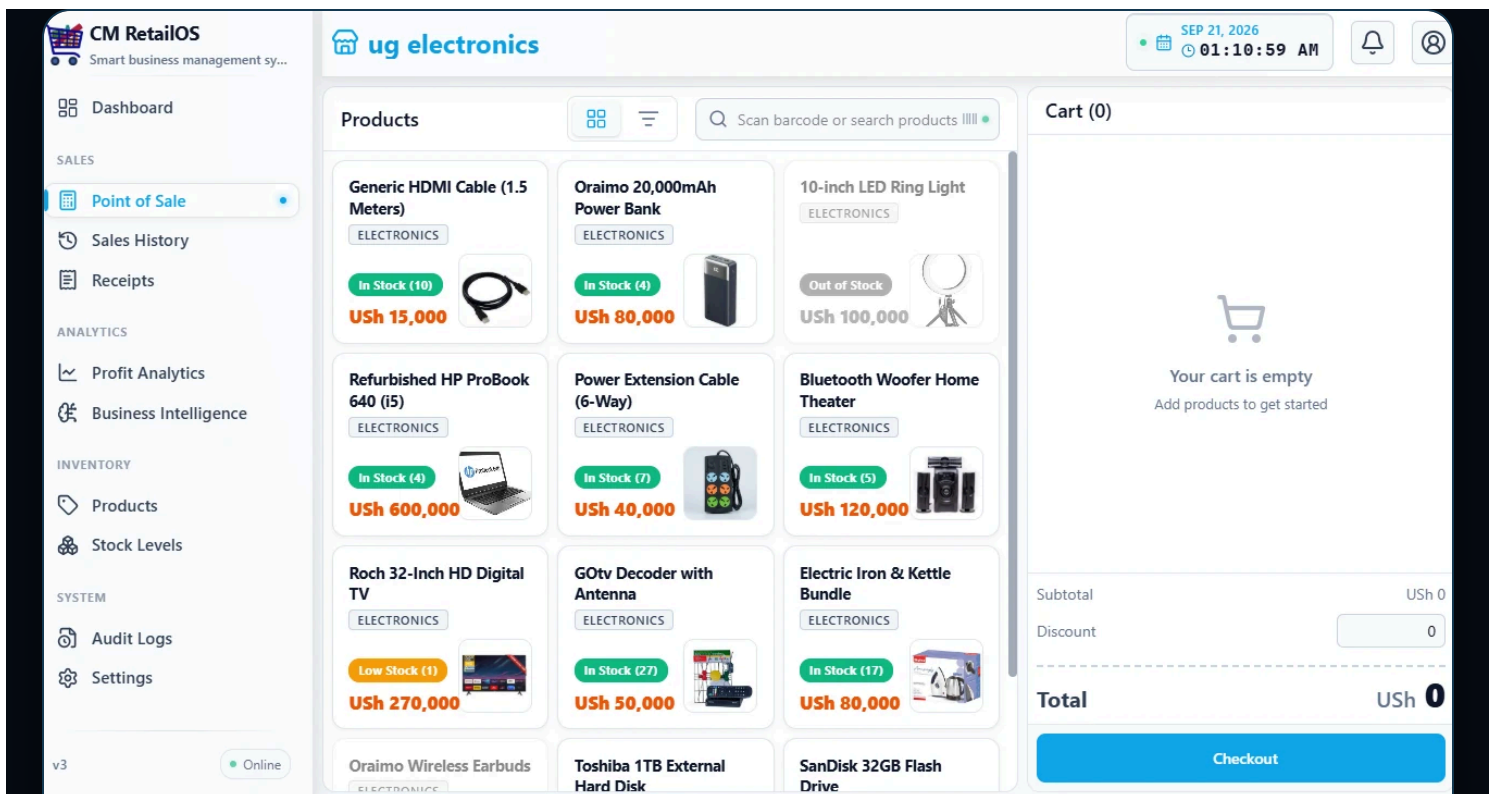


FIGURE 6.2

POS Terminal (Visual Tile Grid Mode) — Product Catalog, Real-Time Stock Badges & Quick Cart

HOW TO READ & USE THIS SCREEN

The POS in Visual Tile Grid mode: high-resolution photo tiles carrying real-time In Stock, Low Stock, and Out of Stock badges on the left, rapid barcode/product search, and the active cart on the right ready to ring up sales and calculate change.

- 1 Tap tiles or scan barcodes to instantly add items to cart — ideal for touchscreens and fast lanes.
- 2 Watch real-time stock badges (In Stock, Low Stock, Out of Stock) while serving customers.
- 3 Apply discounts or adjust line item quantities directly from the cart pane.
- 4 Press Checkout, enter cash tendered, and the thermal receipt prints with change calculated to the shilling.

Pro tip: Out-of-stock tiles are automatically dimmed and unselectable when “Hide Out of Stock” is enabled — the checkout lane always reflects physical inventory reality.

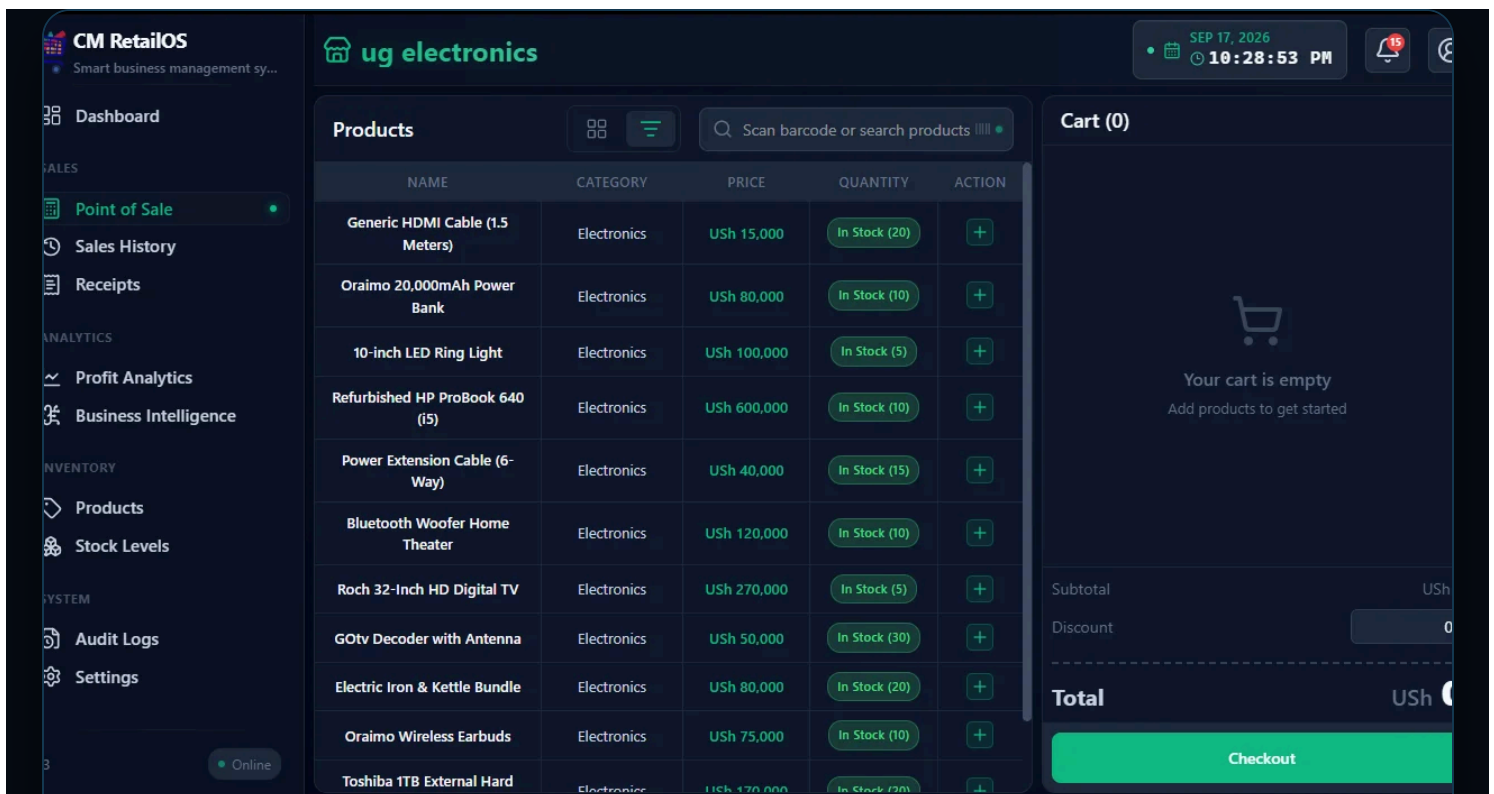


FIGURE 6.3

POS Terminal (Table List View Mode) — Compact Name/Category/Price Rows, One-Tap + Add Buttons, Barcode Scan Box & Cart Ready for the First Scan

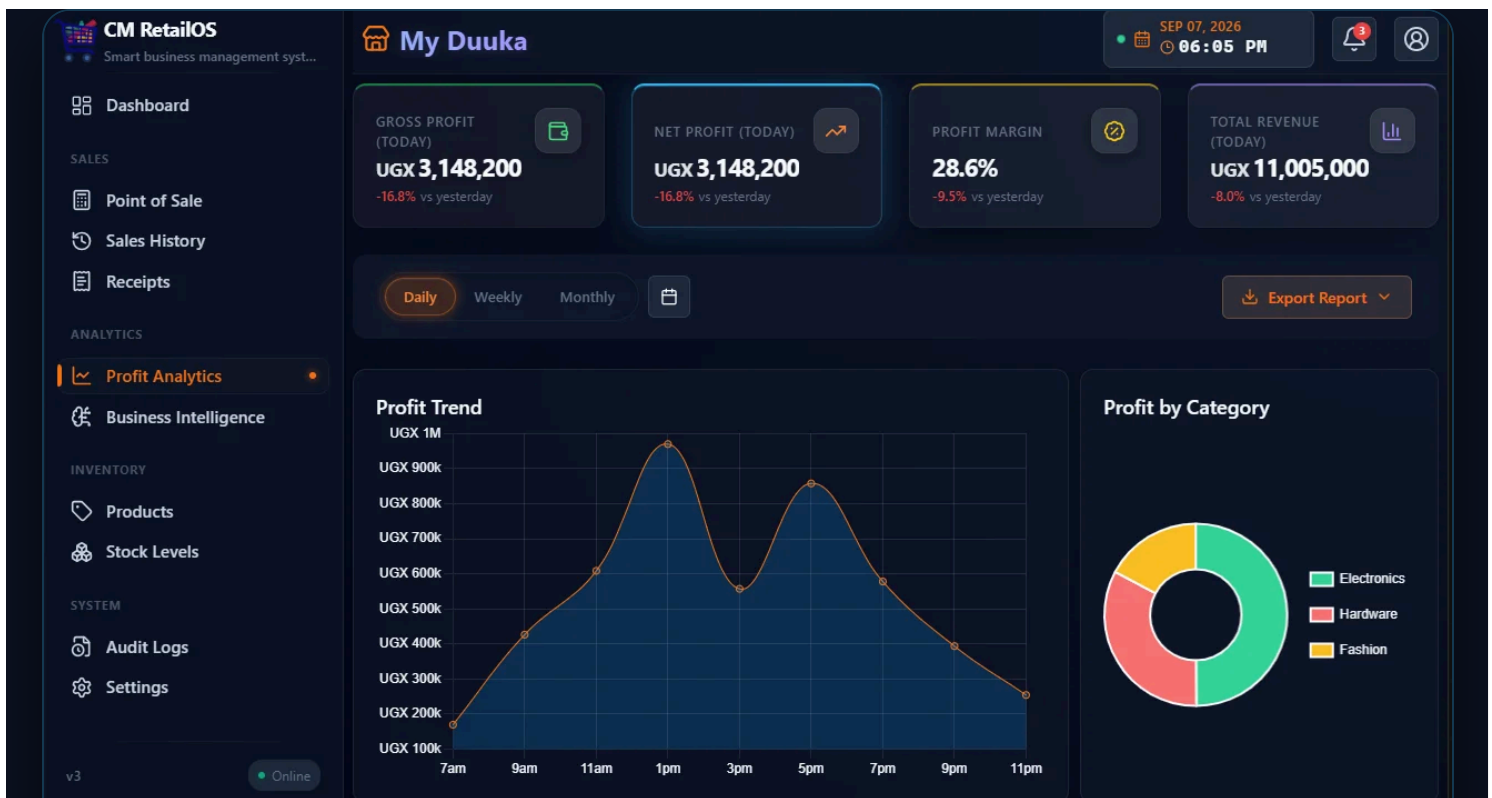
HOW TO READ & USE THIS SCREEN

The same terminal switched to Table List View: compact Name / Category / Price / stock-badge rows with + add buttons, the “Scan barcode or search products” box and an empty cart waiting for the first scan. Table mode suits large baskets and rapid high-volume checkout lanes.

- 1 Toggle between tile and table view with the two view buttons beside the Products title.
- 2 In table mode, scan barcodes rapidly — each beep adds or increments a line.
- 3 Review the cart, apply authorized discounts if needed, then Checkout.

Live Profit Analytics & Business Intelligence

Turn raw checkout transactions into actionable financial intelligence: live net profit margins, peak sales hours, and frequently bought together basket combinations.



HOW TO READ & USE THIS SCREEN

Profit Analytics — the owner's money screen: GROSS PROFIT UGX 3,148,200, NET PROFIT (TODAY), PROFIT MARGIN 28.6% and TOTAL REVENUE UGX 11,005,000, each with vs-yesterday deltas, an hourly Profit Trend curve and the Profit by Category donut (Electronics / Hardware / Fashion), plus an Export Report button.

- 1 Check PROFIT MARGIN daily — a falling margin usually means discounts or price errors.
- 2 Compare gross vs net profit to see what expenses are eating.
- 3 Hover the trend curve to find your most profitable hours.
- 4 Use Export Report to send XLSX/CSV to your accountant or to CM SheetViewer.

⚠ Watch out: This screen is role-restricted. Cashier accounts never see margins or cost prices — keep it that way when assigning permissions.

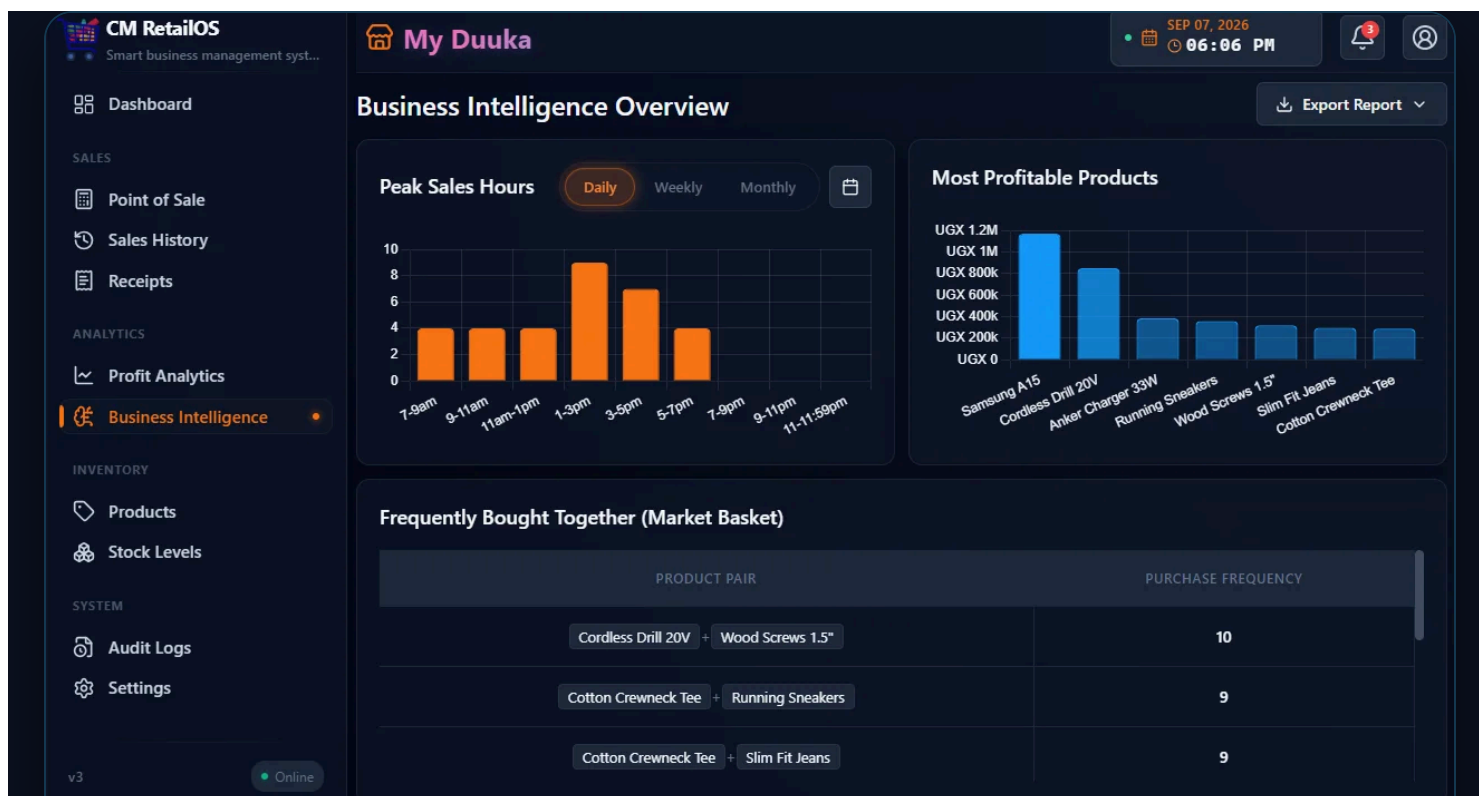


FIGURE 7.2

Business Intelligence Overview — Peak Sales Hours (1-3pm Rush), Most Profitable SKUs & Market Basket Co-Purchase Matrix

HOW TO READ & USE THIS SCREEN

Business Intelligence Overview: the Peak Sales Hours histogram (a clear 1–3 pm rush), the Most Profitable Products ranking, and the Frequently Bought Together market-basket matrix — e.g. Cordless Drill 20V + Wood Screws 1.5", Cotton Crewneck Tee + Running Sneakers — with purchase-frequency bars.

- 1 Schedule your strongest staff and fullest float across the peak hours shown.
- 2 Place frequently-bought-together pairs near each other on shelves to lift basket size.
- 3 Build weekend bundles or promo prices from the top product pairs.
- 4 Review monthly — seasonality shifts both peaks and pairings.

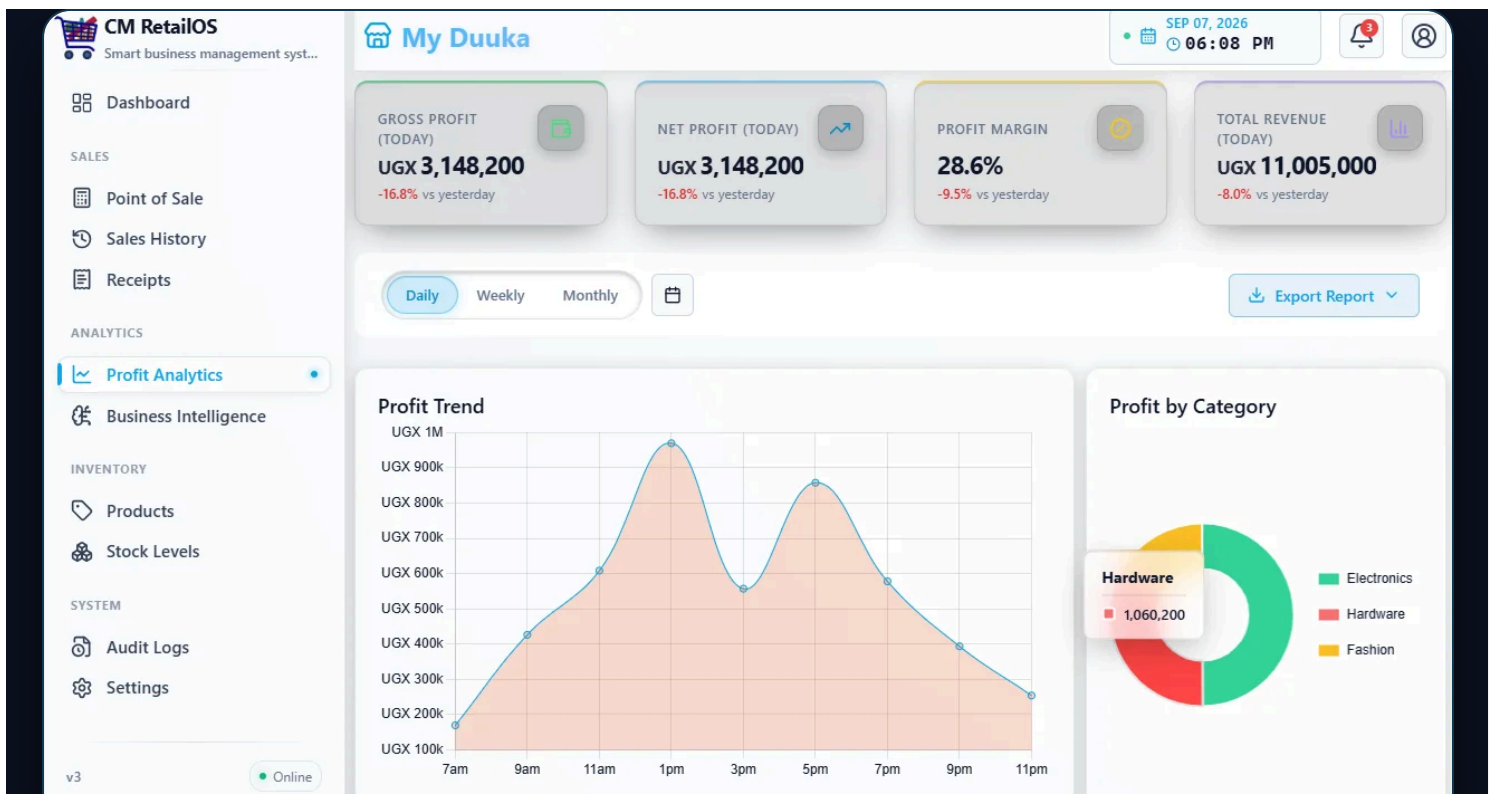


FIGURE 7.3

Profit Analytics (Light Mode) — Category Margin Contribution Tooltip (Hardware: UGX 1.06M)

HOW TO READ & USE THIS SCREEN

Profit Analytics in Light Mode with a category tooltip open: hovering the donut reveals Hardware's exact contribution, UGX 1,060,200. The same data as Figure 7.1, rendered for bright rooms and printed reports.

- 1 Hover each donut segment to read exact profit per category.
- 2 Rank categories by contribution, not just revenue — a quiet category with a fat margin deserves shelf space.
- 3 Switch theme per environment using the appearance controls.

DEDUCTIVE ENGINE

100% ON-DEVICE

ZERO HALLUCINATIONS

Built-in to Profit Analytics & BI Views

7.4 Local Narrative Insights Engine

Most retail POS and inventory platforms either overwhelm business owners with raw numerical grids and static charts, or compromise privacy by uploading sensitive financial records to cloud-based Large Language Models (LLMs).

The CM RetailOS Local Narrative Insights Engine is an on-device, deterministic analytics and natural-language deductive intelligence engine. It runs 100% locally on your computer, transforming raw checkout transactions, inventory buffers, and Cost of Goods Sold (COGS) into clear, actionable, executive-level summaries.

100% Offline & Private

Zero external API calls, zero cloud transmission, and zero risk of proprietary sales leakage. Financial figures remain strictly on your local PC.

Instant & Free

Instantaneous deductions with 0ms network latency. No OpenAI or Anthropic token charges and no recurring monthly SaaS subscriptions.

Zero AI Hallucinations

Grounded directly on deterministic mathematical aggregations and statistical co-occurrence over the local SQLite ledger.

THE 3 CORE DEDUCTIVE TAGS & HOW TO ACT ON THEM

TAG: DEMAND PATTERN

Peak Trading Window Analysis

Mechanism: Bins all completed sales receipts across 9 diurnal trading windows (**7-9am** , **9-11am** , **11am-1pm** , **1-3pm** , **3-5pm** , **5-7pm** , **7-9pm** , **9-11pm** , and **11pm-7am**) to eliminate cashier shift guesswork.

Example Synthesis: *"Your busiest window is 1-3pm. This is the strongest observed sales window for today. Consider aligning coverage and replenishment before it begins. (48 sales)"*

TAG: PROFIT DRIVER

TAG: MARGIN TO REVIEW

Margin Review & Loss Leader Detection

Mechanism: Intersects actual sold quantities with wholesale unit costs (COGS) and discount allowances to reveal true net profit per SKU. High sales volume often blinds retailers to products selling at a loss due to outdated supplier costs or excessive cashier discounting.

Scenario 1 (Profit Leader):

"Cordless Drill 20V leads your profit mix. It is the highest-profit product for this week. Protect its availability and review its margin when pricing changes. (+UGX 840,000 profit)"

Scenario 2 (Silent Margin Leak):

"Basmati Rice 5kg is losing margin. It is the top result for this week, but still below cost. Review its price, cost, or discounting before promoting it. (-UGX 125,000 loss)"

TAG: BASKET OPPORTUNITY

Market Basket Affinity Analysis

Mechanism: A statistical co-occurrence matrix evaluating multi-item basket combinations across completed transactions. It surfaces products frequently purchased together to guide cross-selling, promotional bundling, and shelf adjacencies.

Example Synthesis: "Basmati Rice 5kg and Sunflower Cooking Oil 2L move together. This pairing is a grounded opportunity for a bundle, nearby shelf placement, or checkout suggestion. (34 shared baskets)"

💡 **Executive Dossier Export (.cmp):** When management exports shift or period audits from CM RetailOS, narrative directives are compiled and encrypted with military-grade AES-256 (.cmp format). These confidential executive files can be opened and verified exclusively with the standalone **CM SheetViewer** application.

System Settings, Thermal Printers & Staff RBAC

Customize company profile, thermal receipt templates, POS scanner audio beeps, multi-staff permissions, and 1-click automated database backups.

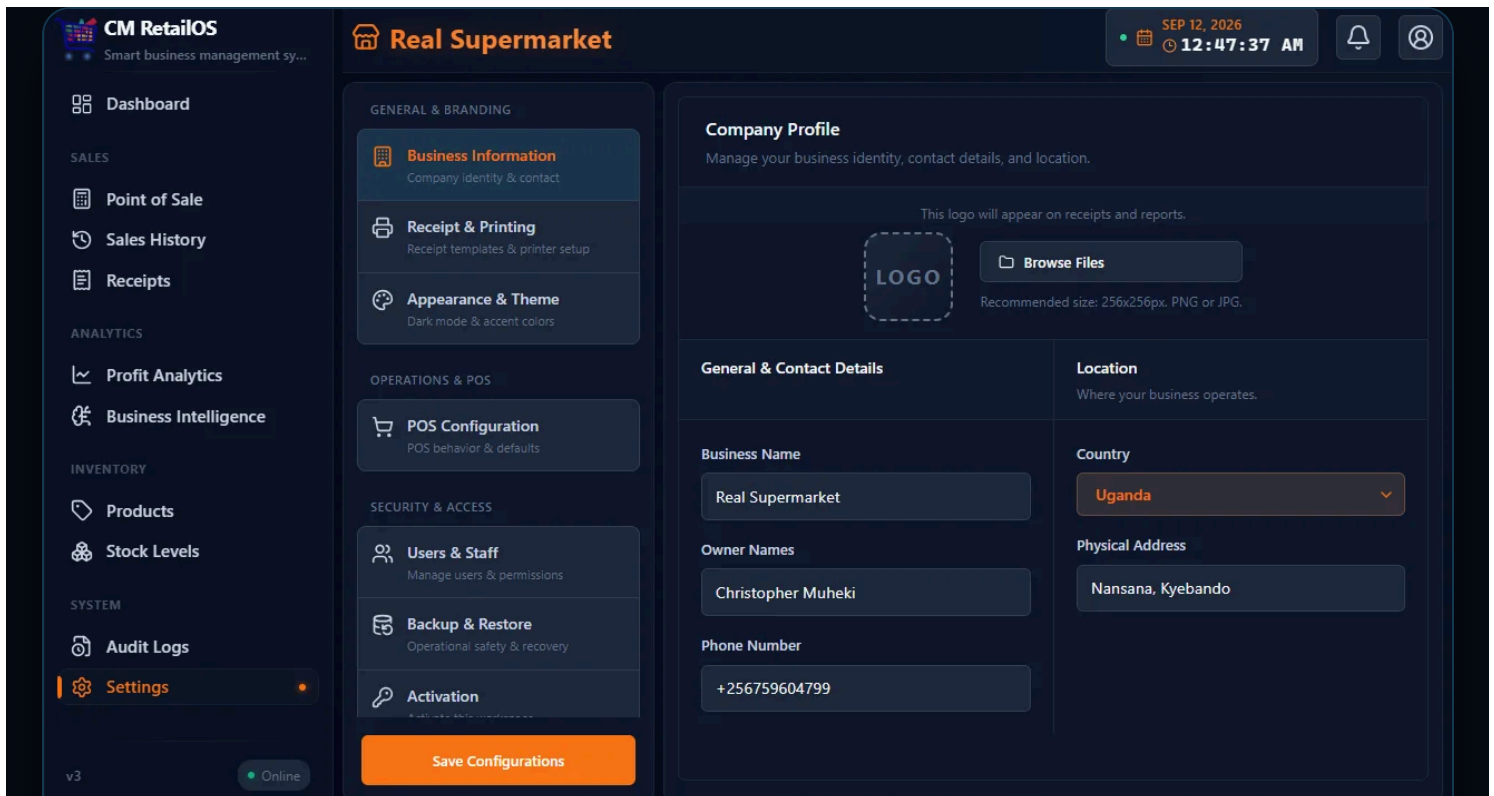


FIGURE 8.1

Settings > Business Information (Store Name, Contact Details & Logo Upload)

HOW TO READ & USE THIS SCREEN

Settings → Business Information: the Company Profile editor with Business Name (Real Supermarket), Owner Names (Christopher Muheki), Phone (+256759604799), a logo uploader (“This logo will appear on receipts and reports”, recommended 256×256 PNG/JPG) and the location block. The left rail groups all settings into General & Branding, Operations & POS and Security & Access.

- 1 Correct any detail that changed — name, phone, address.
- 2 Upload a square logo so receipts and reports carry your brand.
- 3 Press Save Configurations; receipts update from the next sale.

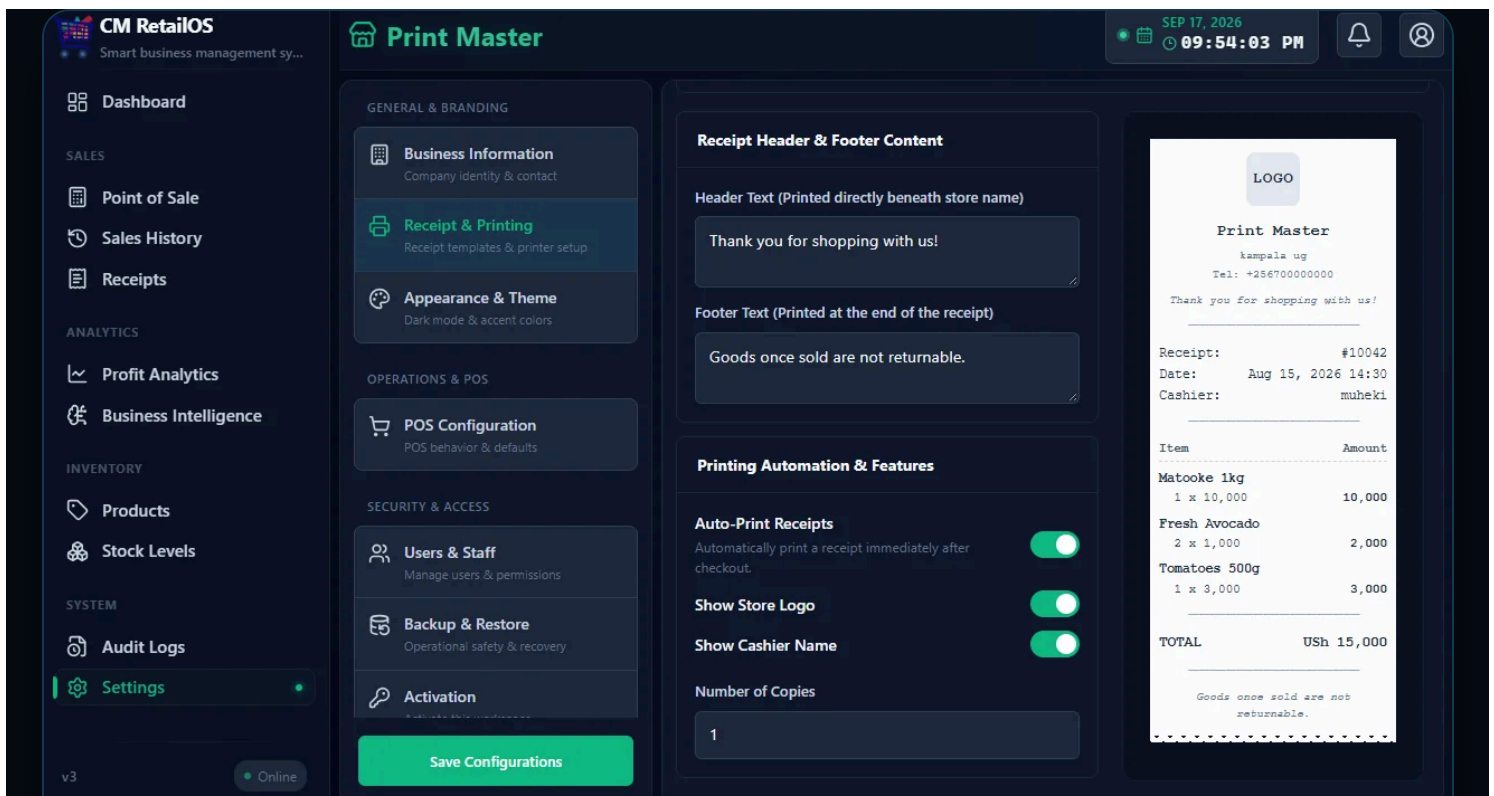


FIGURE 8.2

Settings > Receipt & Printing (Header/Footer Editor & Live Thermal Preview)

HOW TO READ & USE THIS SCREEN

Settings → Receipt & Printing with the live thermal preview on the right: Header Text (“Thank you for shopping with us!”) printed beneath the store name, Footer Text (“Goods once sold are not returnable.”), plus Printing Automation toggles — Auto-Print Receipts, Show Store Logo, Show Cashier Name — and Number of Copies. The preview shows a real slip: Receipt #10042 with Matooke 1kg, Fresh Avocado and Tomatoes totalling USh 15,000.

- 1 Edit header/footer text for promos, helplines or return policies — the preview updates as you type.
- 2 Decide whether the logo and cashier name print on every slip.
- 3 Set copies: 1 for the customer, 2 if you keep a duplicate.
- 4 Keep Auto-Print on for speed at busy counters.

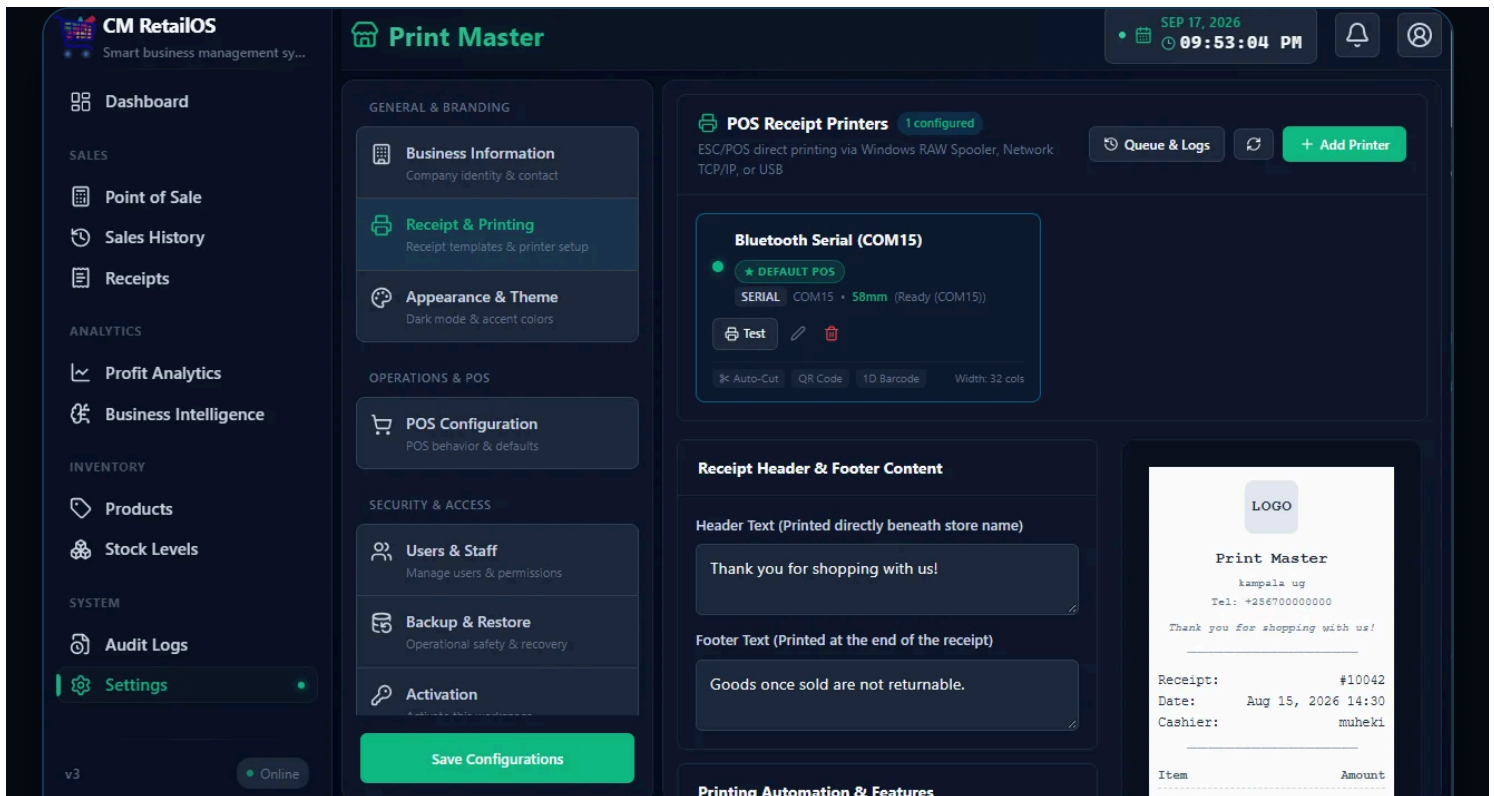


FIGURE 8.3

Settings > Receipt & Printing — Default Printer (EPSON TM-T88V Receipt), Auto-Print Switch, Number of Copies, Show Logo / Cashier Name Toggles

HOW TO READ & USE THIS SCREEN

The printer side of Receipt & Printing: the Default Printer selector (EPSON TM-T88V Receipt in this capture), the Auto-Print Receipts switch and Number of Copies. This is where thermal hardware is bound to the POS.

- 1 Choose your connected thermal printer from the Default Printer list (USB, Bluetooth serial or COM port).
- 2 Match the paper width (58mm/80mm) in the printer's own driver.
- 3 Run a test sale or test print; if nothing prints, check power, paper orientation and the ESC/POS setting.
- 4 Keep one spare paper roll beside the printer at all times.

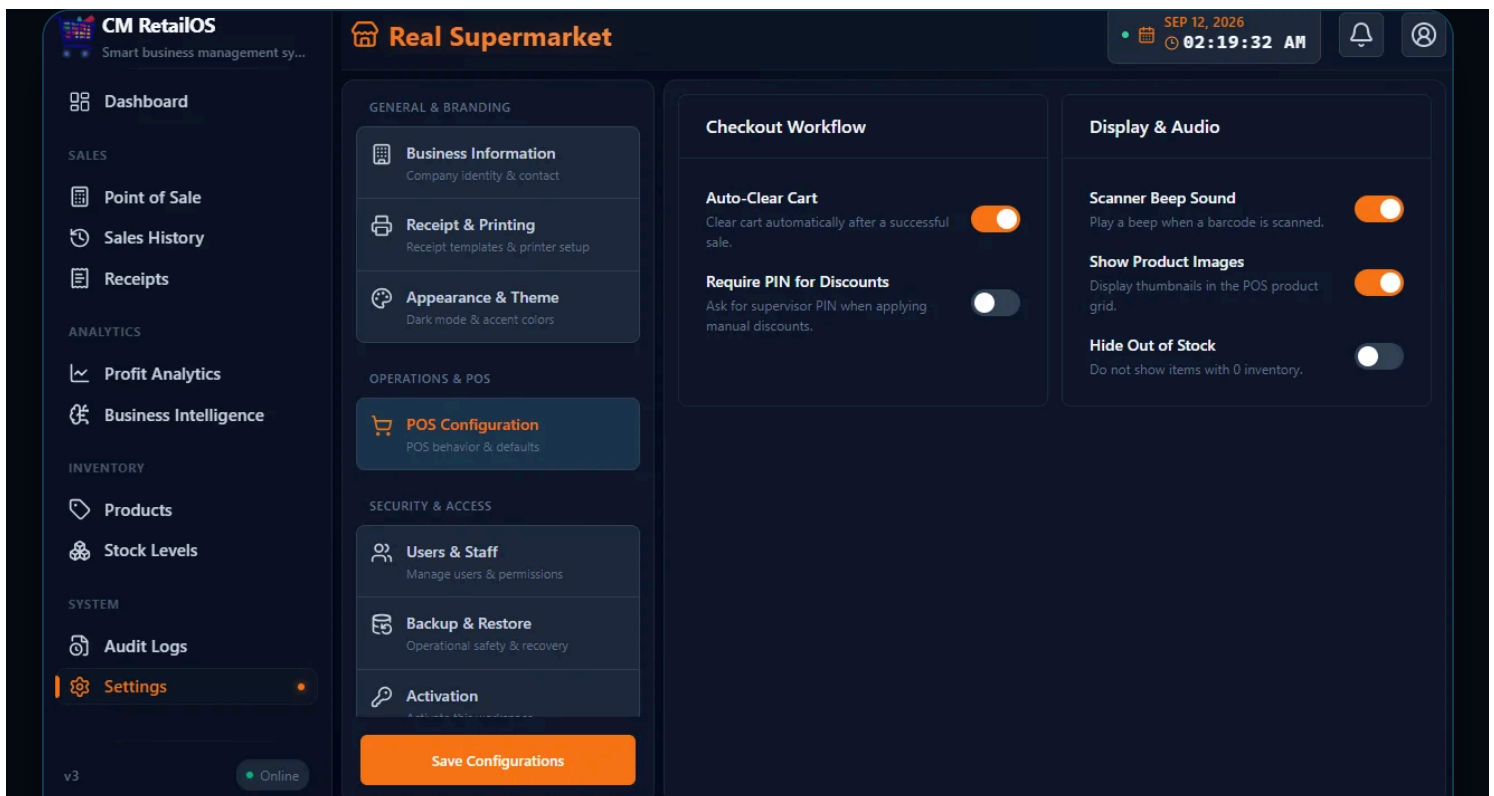


FIGURE 8.4

Settings > POS Configuration (Auto-Clear Cart, Supervisor PIN Discounts & Beep Sounds)

HOW TO READ & USE THIS SCREEN

Settings → POS Configuration — the behaviour switches for the checkout lane: Auto-Clear Cart, Require PIN for Discounts (“Ask for supervisor PIN when applying manual discounts”), and the Display & Audio options: Scanner Beep Sound, Show Product Images, Hide Out of Stock.

- 1 Keep Require PIN for Discounts ON so only supervisors can cut prices.
- 2 Keep Scanner Beep ON — the audible beep is your theft check that every scan registered.
- 3 Enable Hide Out of Stock so cashiers never sell items the shelf cannot fulfil.
- 4 Show Product Images makes tile mode faster for new cashiers.

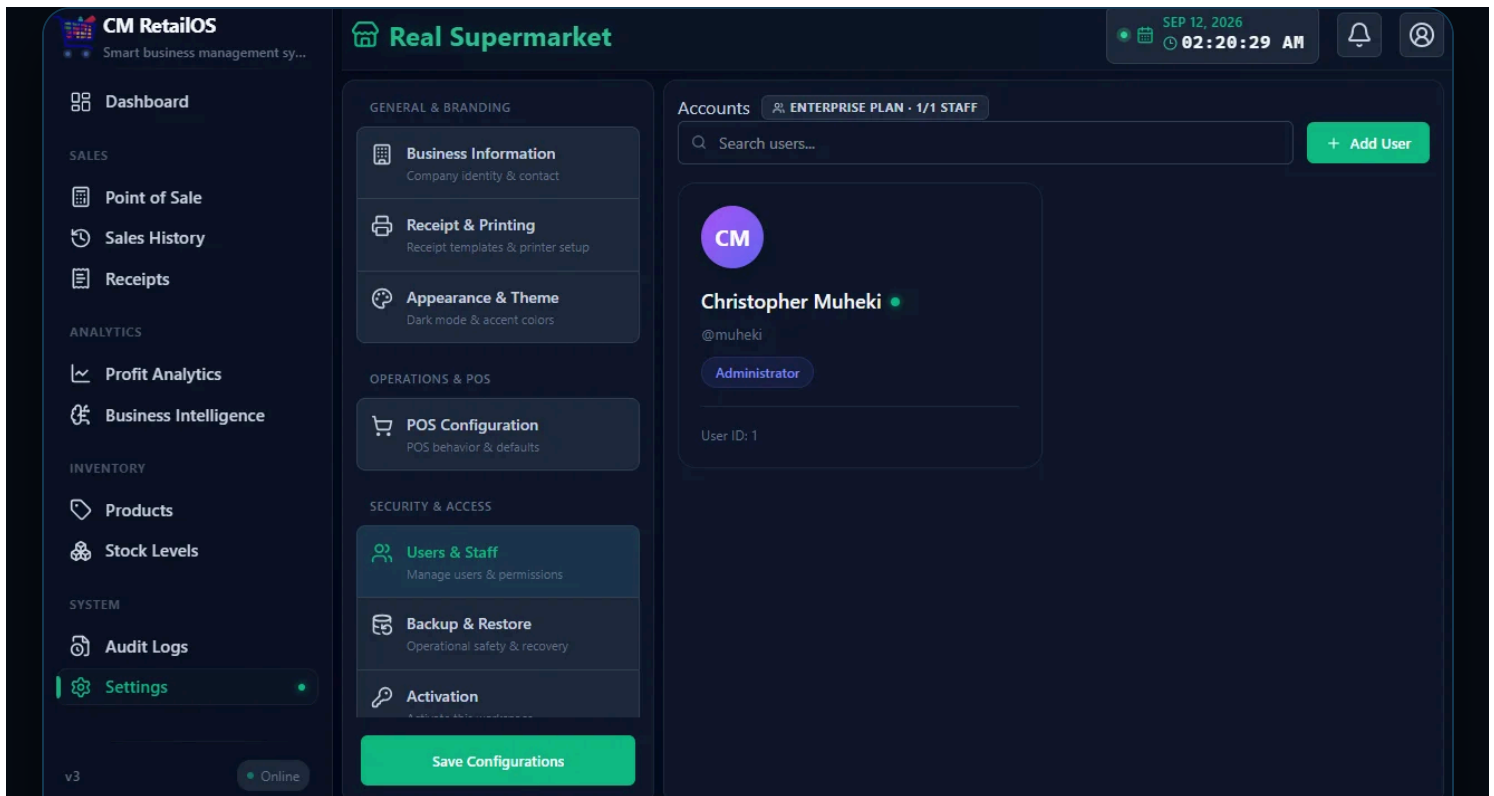


FIGURE 8.5 Users & Staff (1/1 Staff Active - Administrator)

HOW TO READ & USE THIS SCREEN

Settings → Users & Staff on day one: “ENTERPRISE PLAN – 1/1 STAFF” with a single account, Christopher Muheki ([@muheki](#), Administrator, User ID 1). The staff counter shows seats used versus your plan.

- 1 Review who holds the Administrator role — it should be the owner only.
- 2 Use Add User to provision cashiers and managers as you hire.
- 3 Remember: every account created here becomes a sign-in option on the gateway.

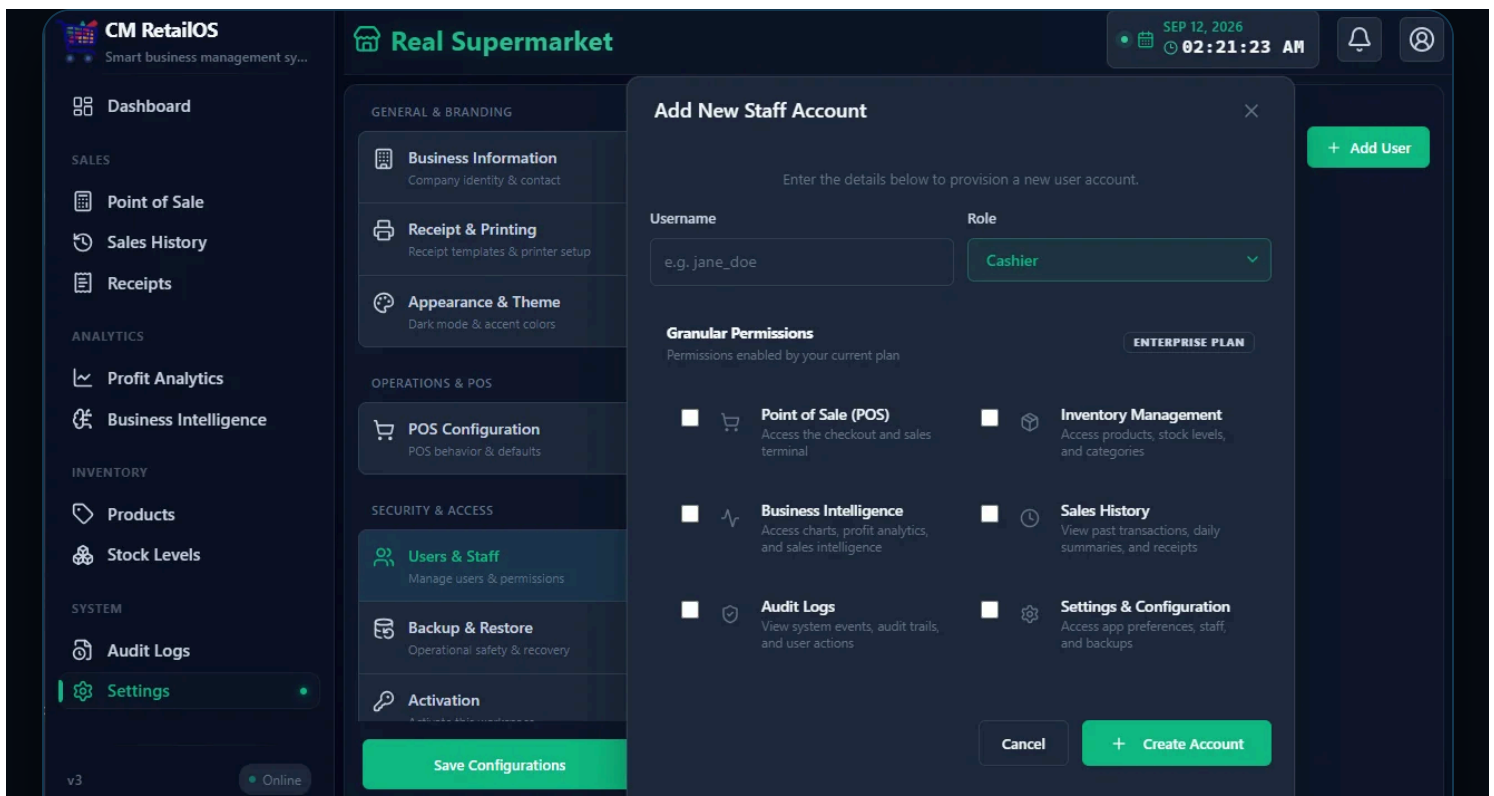


FIGURE 8.6

Add Staff Account — Granular Security Permissions Matrix

HOW TO READ & USE THIS SCREEN

The Add New Staff Account modal: Username field, Role dropdown (Cashier, Manager...) and the Granular Permissions matrix — Point of Sale (POS), Business Intelligence, Inventory Management, Sales History, Settings & Configuration — each with a plain-language description.

- 1 Enter a simple username the staff member will remember.
- 2 Choose the role, then apply least privilege: tick only the permissions the job needs.
- 3 A cashier typically gets POS only; a manager adds Inventory and Sales History.
- 4 Save — cashiers can sign in immediately using just their username; managers use their assigned username and password.

⚠ Watch out: Never tick “Settings & Configuration” for cashiers — it exposes backups, staff management and licence controls.

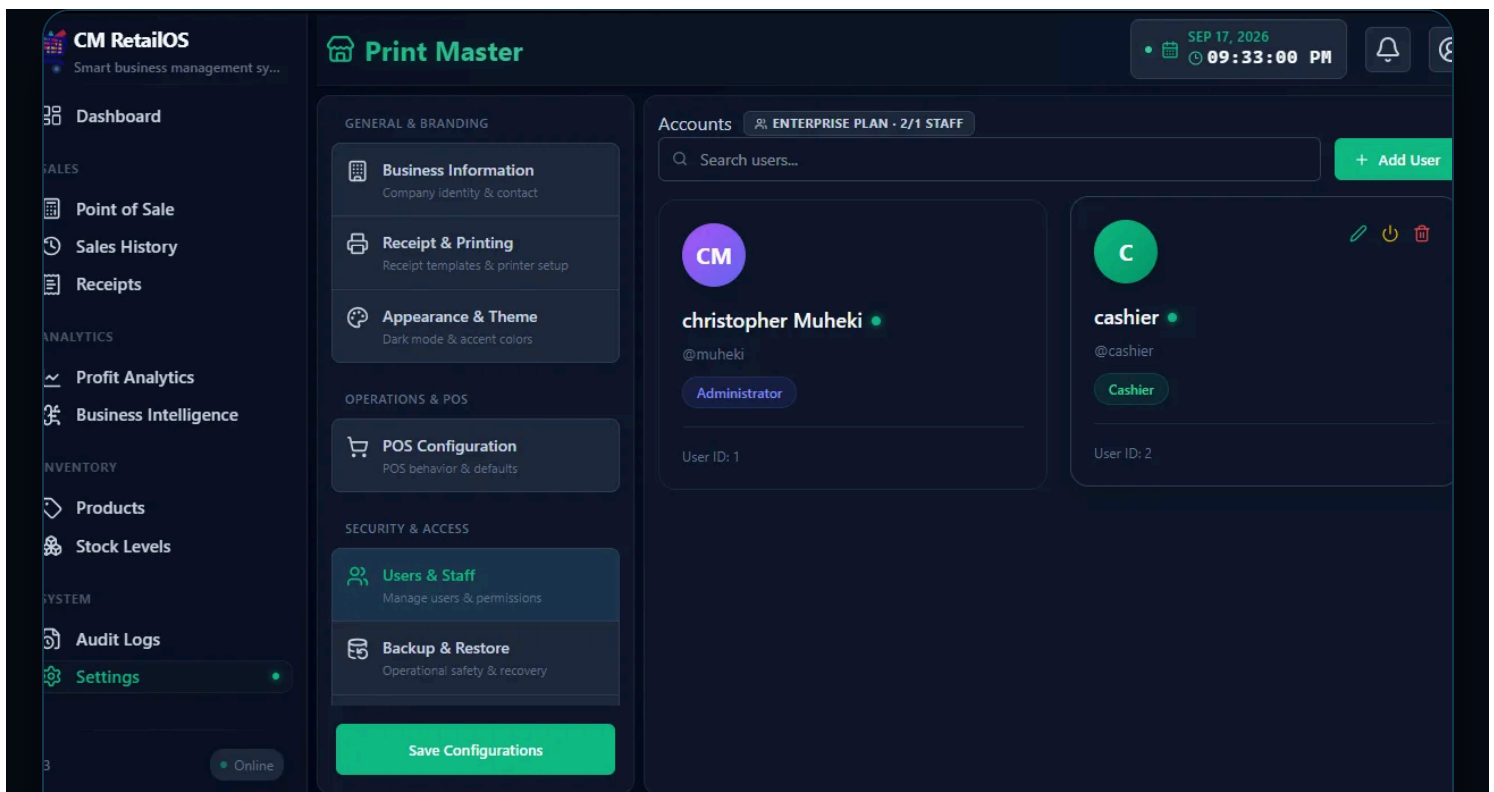


FIGURE 8.7

Users & Staff Directory — Administrator & Cashier Active (Enterprise Plan 2/1 Staff) with Edit, Suspend & Delete Controls per Account

HOW TO READ & USE THIS SCREEN

The grown-up staff directory: “ENTERPRISE PLAN – 2/1 STAFF” listing christopher Muheki (Administrator, User ID 1) and cashier (@cashier, User ID 2). Each account card offers edit (pencil), suspend (power) and delete (trash) controls, with an Add User button and user search above.

- 1 Suspend — don't delete — staff who leave temporarily, keeping their sales history intact.
- 2 Delete only when an account must disappear; past receipts keep their cashier name.
- 3 Use the edit pencil to change roles, permissions, or usernames after staff turnover.

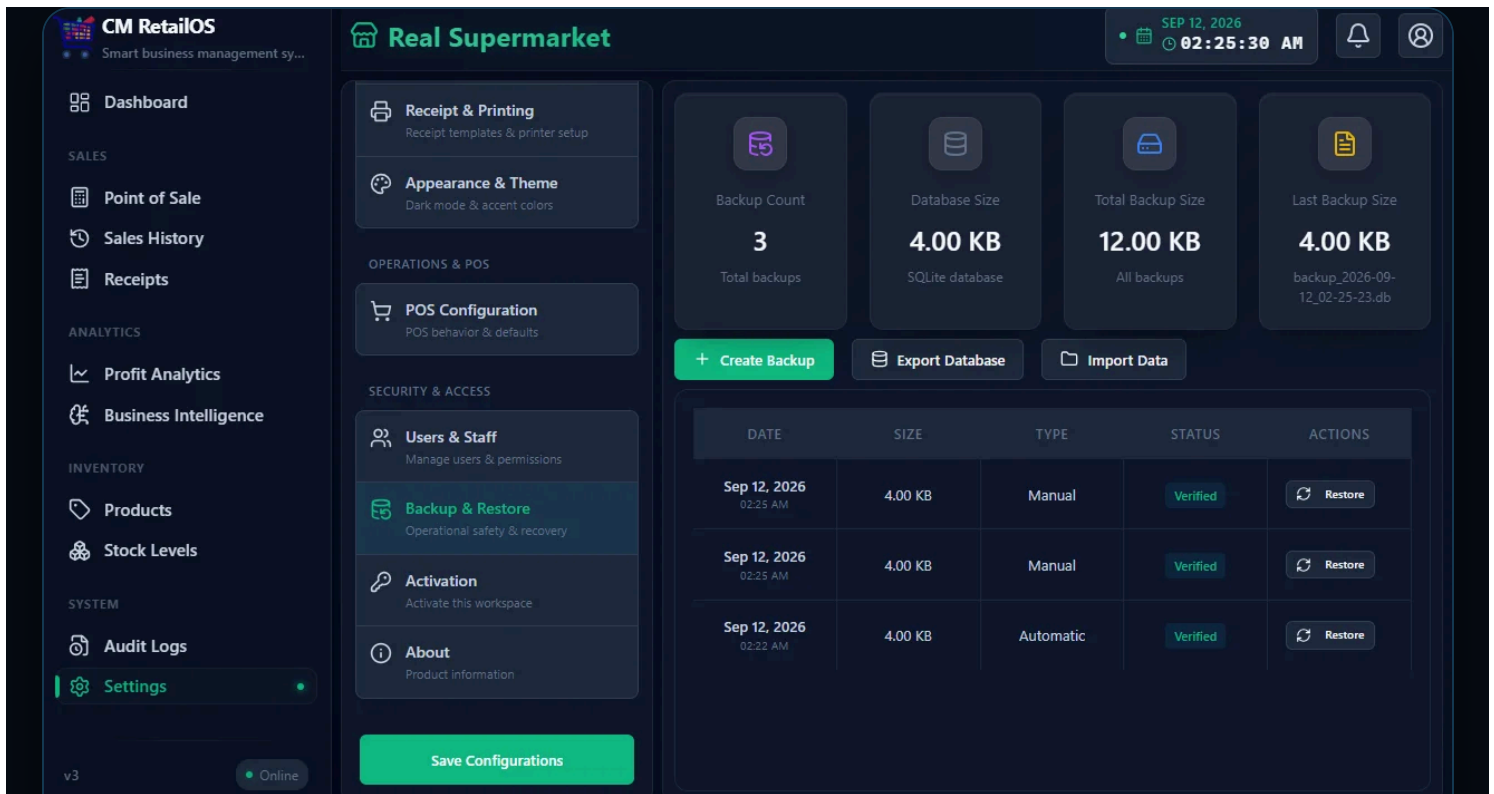


FIGURE 8.8

Settings > Backup & Restore (Automated Snapshots & 1-Click Restore)

HOW TO READ & USE THIS SCREEN

Settings → Backup & Restore: Backup Count 3, Database Size 4.00 KB (SQLite), Last/Total backup sizes, and a table of snapshots with DATE, SIZE, TYPE (Manual/Automatic), STATUS (Verified) and a Restore action per row, plus Export Database and Import Data buttons.

- 1 Let Automatic snapshots run — they capture the database after key events.
- 2 Press Export Database daily and save the file to a USB drive or another computer.
- 3 To recover: choose a verified row and click Restore, or Import Data from an exported file.
- 4 After any restore, check the dashboard totals match your last known figures.

⚠ Watch out: A backup that lives on the same PC as the shop data protects against software faults but not theft or disk death — always keep one copy off the machine.

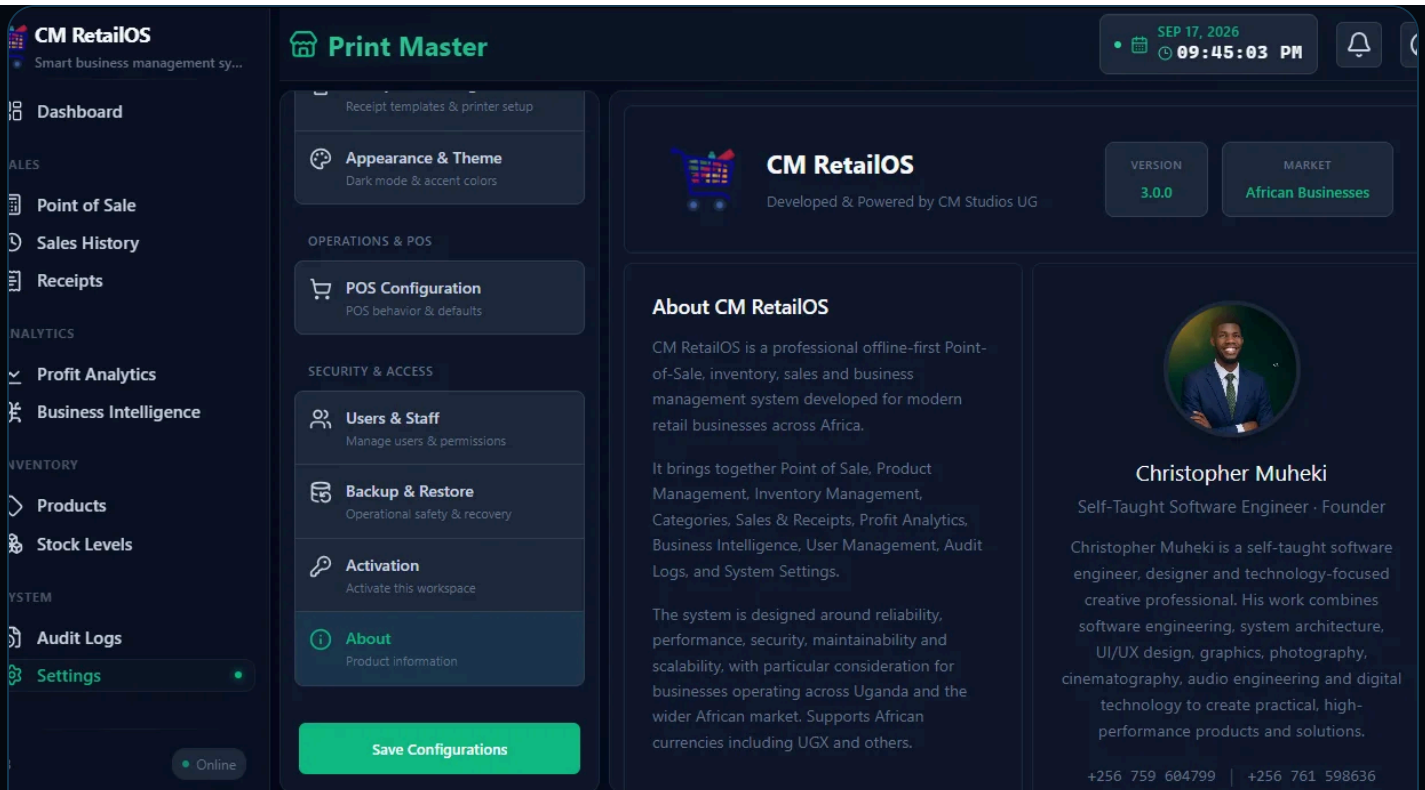


FIGURE 8.9

Settings > About — CM RetailOS v3.0.0 Product Information & Christopher Muheki Developer Profile

HOW TO READ & USE THIS SCREEN

Settings → About: the product identity page — CM RetailOS “Developed & Powered by CM Studios UG”, VERSION 3.0.0, MARKET “African Businesses”, the About CM RetailOS mission text, and the developer profile of Christopher Muheki (Self-Taught Software Engineer · Founder) with support lines +256 759 604799 / +256 761 598636.

- 1 Quote the VERSION number (3.0.0) whenever you contact support.
- 2 Use the listed phone lines for official licence and support requests.
- 3 Scroll the left rail to reach Activation, Backup & Restore and the other system tools.

CM SheetViewer v1.0 Standalone Spreadsheet Engine

Bundled companion desktop application. Open, filter, search, calculate sums, and audit exported Excel (.xlsx, .xls) and CSV sales reports with zero dependencies on Microsoft Office.

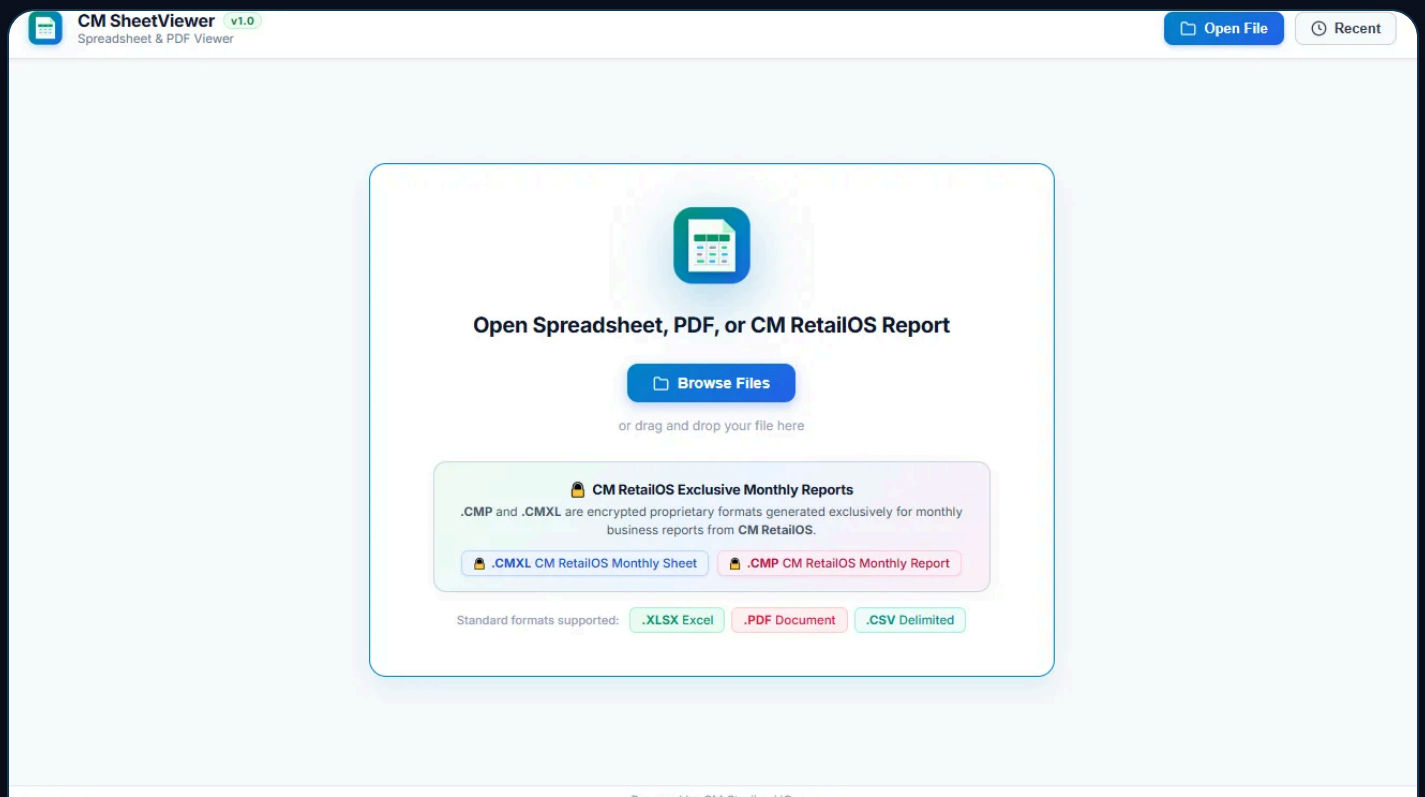


FIGURE 9.1

CM SheetViewer v1.0 — High-Speed Standalone Excel (.xlsx) & CSV Audit Grid Supporting Encrypted **.cmxl** & **.cmp** Monthly Ledgers

HOW TO READ & USE THIS SCREEN

CM SheetViewer v1.0's opening screen — “Open Spreadsheet or PDF Report: Preview merchant exports, inventory catalogs, and financial reports instantly without Microsoft Excel or Acrobat Reader” — with supported formats .XLSX Excel, .CSV Delimited, .PDF Report and .XLSM Macro.

- 1 Export any report from CM RetailOS (Profit Analytics, Sales History, inventory) as XLSX or CSV.
- 2 Open it here with one click — no Office licence, no internet.
- 3 Filter by cashier, transaction status, or date; search rows; compute column sums for quick audits.
- 4 Use it on any Windows PC to review encrypted **.cmxl** / **.cmp** ledgers securely.

 **Pro tip:** SheetViewer is perfect for the accountant's laptop: hand over exported files without ever exposing the live POS database.



How to Audit Reports in CM SheetViewer

COMPANION DESKTOP TOOL

Every time you export sales, inventory, profit sheets, or debtor records from CM RetailOS, you can open them instantly in **CM SheetViewer**:

1. Export Reports from RetailOS

In CM RetailOS, navigate to **Profit Analytics** or **Sales History** and click *Export Report > Excel (.xlsx)* or CSV.

2. One-Click Inspection in SheetViewer

Launch **CM SheetViewer** from your Windows Desktop. Open the exported spreadsheet. The high-speed table viewer renders tens of thousands of rows instantly with zero lag.

Manager Filter & Search Tools: Filter transactions by Cashier Name, receipt number, or date range. Highlight specific categories, calculate column sums, and export print-ready manager audit summaries.

 **CM RetailOS Exclusive Monthly Reports:** .CMP and .CMXL are encrypted proprietary formats **exported automatically** by CM RetailOS for monthly business reports. They strictly require **Admin credentials** to decrypt and open in CM SheetViewer, guaranteeing that confidential store earnings and ledgers can only be accessed by authorized leadership.

Frequently Asked Operations Questions

Quick answers to common daily checkout and management scenarios.

A customer changes their mind about an item after it was scanned. How do I remove it?



Click on the line item in the cart and press the **Delete** key or the red trash icon. If your system is configured with *Cashier Security Lock*, an Admin or Manager must enter their PIN to authorize removing an item from an active cart to prevent theft.

What should the cashier do if the thermal receipt printer runs out of paper mid-sale?



How do I give a discount to a customer without breaking profit reporting?



Can cashiers see our supplier buying prices or our daily profit totals?

